

Active Recall Questions

Mathematics Study Notes

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1 Introduction

Mathematics is not a subject that can be mastered through passive reading alone. True understanding requires the ability to retrieve concepts, definitions, and problem-solving strategies from memory. One of the most effective techniques for achieving this is *active recall*, especially when implemented through a structured list of self-check questions.

1.1 What is Active Recall?

Active recall is the process of deliberately retrieving information from memory rather than re-exposing oneself to it. Instead of rereading notes or textbooks, the learner attempts to answer questions, reconstruct proofs, or solve problems without looking at the solution.

In mathematics, this might include:

- Stating definitions (e.g., convergence of a series),
- Recalling theorems (e.g., the Integral Test),
- Re-deriving formulas,
- Solving representative problems from memory.

1.2 Role of Self-Check Questions

A list of self-check questions provides structure to active recall. These questions act as prompts that guide the learner through the essential components of a topic. For example, in studying infinite series, one might ask:

- What is the definition of convergence?
- When can the Integral Test be applied?
- Why does removing finitely many terms not affect convergence?

Such questions ensure that the learner engages with both conceptual understanding and procedural knowledge.

1.3 Why It Works

Active recall strengthens memory through effortful retrieval. Each time information is successfully recalled, the neural pathways associated with that knowledge become more robust. This leads to:

- Improved long-term retention,
- Faster recall during problem solving,
- Better ability to connect different concepts.

Moreover, attempting to answer questions reveals gaps in understanding. These gaps are often invisible during passive review but become immediately apparent during recall attempts.

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2 Calculus Vol2 - 2. Applications of Integration

<https://openstax.org/books/calculus-volume-2/pages/2-introduction>

2.1 Calculus Vol2 - 2.4 Arc Length of a Curve and Surface Area

<https://openstax.org/books/calculus-volume-2/pages/2-4-arc-length-of-a-curve-and-surface-area>

1. How can the length of a curve be approximated using line segments?
2. How does partitioning an interval lead to a Riemann sum for arc length?
3. What is the formula for the arc length of a curve $y = f(x)$ on $[a, b]$?
4. What assumptions must be satisfied by $f(x)$ for the arc length formula to apply?
5. Why does the arc length formula involve $\sqrt{1 + [f'(x)]^2}$?
6. How is the Pythagorean theorem used in deriving the arc length formula?
7. How do you compute arc length when the curve is given as $x = g(y)$?
8. How does the derivation of arc length for $x = g(y)$ differ from that for $y = f(x)$?
9. What role do smoothness and differentiability play in arc length problems?
10. How can you decide whether to integrate with respect to x or y ?
11. What are common substitutions used when evaluating arc length integrals?
12. How can symmetry simplify arc length computations?
13. What is a surface of revolution?
14. How can a surface of revolution be approximated using frustums of cones?
15. What is a frustum, and why is it useful in this context?
16. How does approximating a surface with frustums lead to a Riemann sum?
17. How is the surface area formula derived from arc length ideas?
18. What is the formula for the surface area when rotating $y = f(x)$ about the x -axis?
19. What is the formula for the surface area when rotating $x = f(y)$ about the y -axis?
20. Why does the surface area formula include a factor of 2π ?
21. What geometric quantity does the factor $2\pi f(x)$ or $2\pi x$ represent?
22. How does arc length appear inside the surface area integral?
23. How do you modify the surface area formula when the curve is given as $x = g(y)$?
24. What similarities exist between arc length and surface area formulas?

2.2 Calculus Vol2 - 2.5 Physical Applications

<https://openstax.org/books/calculus-volume-2/pages/2-5-physical-applications>

1. What is a density function, and how can it vary along a rod or region?
2. How do you compute the mass of a thin rod with variable linear density?
3. What is the formula for mass in terms of an integral over an interval?
4. How does the concept of a Riemann sum apply to mass calculations?
5. How is area density different from linear density?
6. How do you compute the mass of a two-dimensional region with variable density?
7. What is radial density, and in what situations is it used?
8. How does symmetry simplify mass calculations in physical problems?
9. What is work in the context of physics?
10. How is work defined in terms of force and displacement?
11. How can work be approximated using sums of small contributions?
12. What is the general formula for work done by a variable force?
13. How do you compute work when the force depends on position?
14. What is Hooke's Law, and how does it relate force to displacement?
15. How is work computed when stretching or compressing a spring?
16. Why does the integral for spring work involve the force function $F(x) = kx$?
17. How do you set up limits of integration in spring problems?
18. What is a pumping problem in calculus?
19. How is work computed when lifting a fluid (e.g., water) from a tank?
20. How do you model a fluid as a collection of thin horizontal slices?
21. What quantities must be determined for each slice in a pumping problem?
22. How do you determine the distance each slice must be lifted?
23. How does density of a fluid affect the work calculation?
24. What is the role of weight density in fluid problems?
25. How do you express the work required to pump fluid as an integral?
26. What is hydrostatic pressure?
27. How is pressure related to depth in a fluid?
28. What units are used for pressure?
29. How do you compute the force exerted by a fluid on a submerged surface?
30. How is force related to pressure and area?
31. How do you model a submerged surface using horizontal strips?
32. How do you compute the force on each strip?
33. How is the total force found using an integral?
34. What assumptions are made about fluid density in hydrostatic problems?
35. How can changing the shape of a container affect work or force calculations?

2.3 Calculus Vol2 - 2.6 Moments and Centers of Mass

<https://openstax.org/books/calculus-volume-2/pages/2-6-moments-and-centers-of-mass>

1. What is the definition of the center of mass of a system?
2. How can the center of mass be interpreted physically as a “balancing point”?
3. What is a moment of a system of masses with respect to a point (e.g., the origin)?
4. For point masses on a line, how do you compute the total mass?
5. For point masses on a line, how do you compute the moment with respect to the origin?
6. How do you compute the center of mass $\bar{x}$ for discrete point masses on a line?
7. How does the formula for center of mass relate to a weighted average?
8. For point masses in the plane, how are the moments M_x and M_y defined?
9. How do you compute the coordinates $(\bar{x}, \bar{y})$ of the center of mass in the plane?
10. Why does the center of mass depend more heavily on regions with larger mass?
11. What changes when moving from discrete masses to a continuous mass distribution?
12. What is a lamina?
13. What assumption is often made about density when studying laminas?
14. How is mass computed for a continuous distribution along a line?
15. How is mass computed for a lamina bounded by a function $f(x)$ on $[a, b]$?
16. How do you define the moment of a continuous mass distribution using integrals?
17. How are M_x and M_y computed for a lamina using integrals?
18. How do you compute the center of mass $(\bar{x}, \bar{y})$ for a lamina?
19. What is the centroid of a region?
20. When does the centroid coincide with the center of mass?
21. Why does constant density simplify center of mass calculations?
22. How can symmetry be used to determine the center of mass without integration?
23. Where is the center of mass of a rectangular lamina located?
24. What role does partitioning (Riemann sums) play in deriving formulas for mass and moments?
25. How do integrals arise naturally from approximating a lamina by small rectangles?
26. What is the geometric meaning of M_x and M_y ?
27. Why is the center of mass independent of density when density is constant?
28. How would variable density change the formulas for mass and moments?
29. How does the concept of center of mass connect to real-world applications (e.g., balance, engineering)?
30. What is the relationship between center of mass and equilibrium?
31. How can you check whether your computed center of mass is reasonable using symmetry or intuition?
32. How does the idea of “balancing point” generalize from 1D to 2D regions?
33. What are the key steps to solve a typical center of mass problem for a region?
34. What are the key differences between discrete and continuous mass systems?
35. How can you set up integrals for regions bounded by curves when finding centers of mass?

2.4 Calculus Vol2 - 2.7 Integrals, Exponential Functions, and Logarithms

<https://openstax.org/books/calculus-volume-2/pages/2-7-integrals-exponential-functions-and-logarithms>

1. How can the natural logarithm function $\ln x$ be defined using a definite integral?
2. Why does the usual power rule for integration fail for $\int \frac{1}{x}, dx$, and how does the integral definition of $\ln x$ resolve this?
3. What is the domain of the function $\ln x$ when defined via an integral, and why?
4. How is $\ln x = \int_1^x \frac{1}{t}, dt$ interpreted geometrically?
5. What is the value of $\ln 1$, and how does it follow from the integral definition?
6. Is $\ln x$ increasing or decreasing on its domain? How can this be seen from the integral definition?
7. What is the derivative of $\ln x$, and how does it follow from the Fundamental Theorem of Calculus?
8. How can you use the definition of $\ln x$ to show it is continuous and differentiable?
9. How is the number e defined using the natural logarithm?
10. What special property defines the number e in terms of $\ln x$?
11. How is the exponential function e^x defined as an inverse function?
12. What is the derivative of e^x , and why is this property significant?
13. What is the integral of e^x ?
14. How do you compute $\int e^{kx}, dx$ for a constant k ?
15. How are general exponential functions a^x expressed in terms of e^x ?
16. What is the formula for $\int a^x, dx$ and how does $\ln a$ appear in it?
17. How are general logarithmic functions $\log_a x$ expressed using $\ln x$?
18. What is the derivative of $\log_a x$ in terms of $\ln a$?
19. What are the key logarithmic properties (product, quotient, power rules), and how can they be justified using integrals?
20. How can you prove that $\ln(ab) = \ln a + \ln b$ using the integral definition?
21. How can you prove that $\ln(a^r) = r \ln a$?

3 Calculus Vol2 - 4. Introduction to Differential Equations

<https://openstax.org/books/calculus-volume-2/pages/4-introduction>

3.1 Calculus Vol2 - 4.1 Basics of Differential Equations

<https://openstax.org/books/calculus-volume-2/pages/4-1-basics-of-differential-equations>

1. What is a differential equation?
2. What is meant by a solution to a differential equation?
3. How can you verify that a function $y = f(x)$ is a solution to a given differential equation?
4. Give an example of a differential equation and one of its solutions.
5. Why are solutions to differential equations often not unique?
6. What property of derivatives explains the non-uniqueness of solutions?
7. What is the order of a differential equation?
8. How do you determine the order of a differential equation from its expression?
9. What is the order of the differential equation $y' - 4y = x^2 - 3x + 4$?
10. What is the order of the differential equation $x^2y''' - 3xy'' + xy' - 3y = \sin x$?
11. What is a general solution of a differential equation?
12. Why does a general solution typically include an arbitrary constant C ?
13. For the differential equation $y' = 2x$, what is its general solution?
14. What does the family of functions $y = x^2 + C$ represent?
15. What is a particular solution of a differential equation?
16. How do you obtain a particular solution from a general solution?
17. What is an initial-value problem?
18. How does an initial condition help determine a unique solution?
19. Solve the initial-value problem:
$$y' = 2x, \quad y(2) = 7.$$
20. What is the difference between a general solution and a particular solution?
21. What does it mean for a function to satisfy a differential equation?
22. Verify that $y = e^{-3x} + 2x + 3$ is a solution of
$$y' + 3y = 6x + 11.$$
23. Verify that $y = 2e^{3x} - 2x - 2$ is a solution of
$$y' - 3y = 6x + 4.$$
24. When checking a solution, why must both y and its derivatives be substituted into the equation?
25. Can two different functions be solutions to the same differential equation? Explain why.
26. What role does the constant C play geometrically in the family of solutions?
27. How would the graph of $y = x^2 + C$ change as C varies?
28. Why is identifying the order of a differential equation useful?
29. What information is needed in addition to a differential equation to determine a unique solution?

3.2 Calculus Vol2 - 4.2 Direction Fields and Numerical Methods

<https://openstax.org/books/calculus-volume-2/pages/4-2-direction-fields-and-numerical-methods>

1. What is a direction field (slope field) for a differential equation?
2. For what type of differential equations are direction fields typically used?
3. What is the general form of a first-order differential equation used in direction fields?
4. What does each small line segment in a direction field represent?
5. How is the slope of a line segment at a point (x, y) determined?
6. How do you construct a direction field for a differential equation $y' = f(x, y)$?
7. Why can a direction field be drawn without solving the differential equation explicitly?
8. What qualitative information can a direction field provide about solutions?
9. What is a solution curve in the context of a direction field?
10. How can you sketch a solution curve using a direction field?
11. Why do solution curves follow the line segments in a direction field?
12. What is an initial condition in the context of direction fields?
13. How do you use an initial point (x_0, y_0) to sketch a particular solution?
14. Why do solution curves corresponding to different initial conditions typically not intersect?
15. What is an equilibrium solution?
16. How can equilibrium solutions be identified from a differential equation?
17. How do equilibrium solutions appear in a direction field?
18. What does it mean for an equilibrium solution to be stable?
19. What does it mean for an equilibrium solution to be unstable?
20. How can you determine stability visually from a direction field?
21. What is the purpose of numerical methods in differential equations?
22. Why are numerical methods needed when solving differential equations?
23. What is Euler's Method?
24. What type of problems is Euler's Method used to approximate?
25. What is the idea behind Euler's Method in terms of tangent line approximation?
26. What is the step size h in Euler's Method?
27. How does the choice of step size affect the accuracy of Euler's Method?
28. What is the iterative formula for Euler's Method?
29. Starting from (x_n, y_n) , how do you compute (x_{n+1}, y_{n+1}) ?
30. Write the update rule:
$$y_{n+1} = y_n + hf(x_n, y_n).$$
31. How is x_{n+1} computed from x_n ?

32. Apply Euler's Method to approximate the solution of

$$y' = f(x, y), \quad y(x_0) = y_0$$

for one step.

33. Apply Euler's Method for two steps and express the result explicitly.

34. What geometric idea explains why Euler's Method works?

35. Why does Euler's Method accumulate error over multiple steps?

36. What is the difference between the exact solution and a numerical approximation?

37. How can decreasing the step size h improve the approximation?

38. What is the trade-off when choosing a very small step size?

39. How is Euler's Method related to linear approximation (tangent lines)?

40. In what sense does Euler's Method "follow" the direction field?

41. How would you approximate a solution curve using only a direction field (without formulas)?

42. What are the limitations of direction fields?

43. What are the limitations of Euler's Method?

44. When would you prefer a graphical method over an analytical solution?

45. When would you prefer a numerical method over solving analytically?

46. How can direction fields and Euler's Method be used together?

3.3 Calculus Vol2 - 4.3 Separable Equations

<https://openstax.org/books/calculus-volume-2/pages/4-3-separable-equations>

1. What is a separable differential equation?
2. In what form must a differential equation be written to be considered separable?
3. Write the general form:

$$\frac{dy}{dx} = f(x)g(y).$$

4. Why is such an equation called “separable”?
5. What does it mean to separate variables in a differential equation?
6. How do you rewrite a separable differential equation in differential form?
7. Write the separated form:

$$\frac{dy}{g(y)} = f(x) dx.$$

8. What is the main idea behind the method of separation of variables?
9. What are the steps of the separation of variables method?
10. Why must you check for values where $g(y) = 0$ before separating variables?
11. What do solutions of $g(y) = 0$ represent?
12. Why can constant solutions be lost during the separation process?
13. After separating variables, what is the next step?
14. Why do we integrate both sides of the equation?
15. What form does the equation take after integration?
16. Why is an arbitrary constant C introduced after integration?
17. When solving, why might it not be possible to express y explicitly as a function of x ?
18. What is an implicit solution?
19. Give an example of an implicit solution arising from separation of variables.
20. How do you apply an initial condition to a separable differential equation?
21. What is an initial-value problem in this context?
22. After integrating, how do you solve for the constant using an initial condition?
23. Solve the general structure:

$$\int \frac{1}{g(y)} dy = \int f(x) dx.$$

24. For the equation $y' = (x^2 - 4)(3y + 2)$, how do you separate variables?
25. What constant solution arises from $3y + 2 = 0$?
26. After separation, what integral must be computed:

$$\int \frac{dy}{3y + 2}?$$

27. What substitution is useful for integrating expressions like $\frac{1}{3y+2}$?
28. Why is substitution often needed when integrating the y -side?

29. How do you interpret the final solution after integrating both sides?
30. For the equation $y' = (2x + 3)(y^2 - 4)$, how do you separate variables?
31. What constant solutions arise from $y^2 - 4 = 0$?
32. Why do these constant solutions need to be included separately?
33. What integration technique is typically used for expressions like $\frac{1}{y^2-4}$?
34. How does partial fraction decomposition appear in separable equations?
35. Why are integration techniques from earlier calculus essential here?
36. What is an autonomous differential equation?
37. How can you recognize an autonomous equation from its form?
38. Why is an equation of the form $y' = g(y)$ separable?
39. What is the general solution to an equation of the form $y' = f(x)$?
40. What is the general solution to an equation of the form $y' = g(y)$?
41. How does separation of variables generalize these simpler cases?
42. In what situations is separation of variables especially useful?
43. Why are separable equations common in physics and engineering?
44. What are the limitations of the separation of variables method?
45. Can every differential equation be solved using separation of variables?
46. How can you recognize when a differential equation is not separable?
47. What is the overall strategy when solving separable differential equations?

3.4 Calculus Vol2 - 4.4 The Logistic Equation

<https://openstax.org/books/calculus-volume-2/pages/4-4-the-logistic-equation>

1. What real-world limitation motivates the logistic model of population growth?
2. Why is exponential growth not realistic for large populations?
3. What is the carrying capacity of a population?
4. What does the symbol K represent in the logistic equation?
5. What does the symbol r represent in the logistic equation?
6. Write the logistic differential equation:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right).$$

7. Why is the logistic equation nonlinear?
8. What happens to $\frac{dP}{dt}$ when P is very small?
9. How does the logistic equation behave when $P \ll K$?
10. How does the logistic model approximate exponential growth for small populations?
11. What happens to $\frac{dP}{dt}$ when $P = K$?
12. What does it mean physically when $P = K$?
13. What happens to $\frac{dP}{dt}$ when $P > K$?
14. Why does the population decrease when $P > K$?
15. What are the equilibrium solutions of the logistic equation?
16. Solve:

$$rP \left(1 - \frac{P}{K} \right) = 0.$$

17. Why are $P = 0$ and $P = K$ equilibrium solutions?
18. What is meant by stability of an equilibrium solution?
19. Is $P = 0$ stable, unstable, or semi-stable? Explain.
20. Is $P = K$ stable, unstable, or semi-stable? Explain.
21. What is a phase line?
22. How do you construct a phase line for the logistic equation?
23. What does the phase line indicate about the sign of $\frac{dP}{dt}$?
24. For which values of P is the population increasing?
25. For which values of P is the population decreasing?
26. How does the phase line describe long-term behavior of solutions?
27. What happens to $P(t)$ as $t \rightarrow \infty$ when $0 < P_0 < K$?
28. What happens to $P(t)$ as $t \rightarrow \infty$ when $P_0 > K$?
29. Why does the solution approach K but never reach it exactly?
30. What is the general strategy to solve the logistic differential equation?
31. Why is the logistic equation separable?

32. Rewrite the logistic equation in separated form.
33. What algebraic manipulation is needed before integrating?
34. Why is partial fraction decomposition used in solving the logistic equation?
35. What is the general solution of the logistic differential equation?
36. Write the solution:

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}.$$

37. What is the role of the initial condition $P(0) = P_0$?
38. How does the solution behave as $t \rightarrow \infty$?
39. How does the solution behave as $t \rightarrow -\infty$?
40. What is the shape of the graph of a logistic function?
41. Why is the logistic curve called sigmoidal (S-shaped)?
42. What is a point of inflection?
43. Why does the logistic solution have a point of inflection?
44. At what population value does the inflection point occur?
45. Why is the growth rate maximal at $P = \frac{K}{2}$?
46. How can you find the inflection point using the second derivative?
47. What happens to concavity before and after the inflection point?
48. How does the logistic model improve on exponential growth models?
49. In what types of real-world systems is the logistic equation used?
50. What assumptions underlie the logistic model?
51. What are the limitations of the logistic equation?

3.5 Calculus Vol2 - 4.5 First-order Linear Equations

<https://openstax.org/books/calculus-volume-2/pages/4-5-first-order-linear-equations>

1. What is a first-order linear differential equation?
2. What is the general (standard) form of a first-order linear differential equation?
3. Write the standard form:

$$y' + p(x)y = q(x).$$

4. How can any first-order linear differential equation be rewritten into standard form?
5. What roles do the functions $p(x)$ and $q(x)$ play?
6. Why is the equation called “linear”?
7. What distinguishes a linear differential equation from a separable one?
8. Can every linear differential equation be solved by separation of variables? Why or why not?
9. What is an integrating factor?
10. Why is an integrating factor useful when solving linear differential equations?
11. What property do we want the integrating factor to create?
12. What is the formula for the integrating factor $\mu(x)$?
13. Write:

$$\mu(x) = e^{\int p(x) dx}.$$

14. How is the integrating factor derived?
 15. What differential equation does $\mu(x)$ satisfy?
 16. Write:
- $$\mu'(x) = \mu(x)p(x).$$
17. After multiplying by $\mu(x)$, what form does the equation take?
 18. Why does the left-hand side become a derivative of a product?
 19. Write the identity:

$$\frac{d}{dx}(\mu(x)y) = \mu(x)y' + \mu(x)p(x)y.$$

20. After multiplying by $\mu(x)$, what equation do we obtain?
21. Write:

$$\frac{d}{dx}(\mu(x)y) = \mu(x)q(x).$$

22. What is the next step after obtaining this form?
23. Why can both sides now be integrated easily?
24. After integrating, what form does the solution take?
25. Write the general solution:

$$\mu(x)y = \int \mu(x)q(x) dx + C.$$

26. How do you solve for $y(x)$ after integration?
27. What is the full step-by-step strategy for solving first-order linear equations?
28. List the five steps of the method.

29. Why is it important to first rewrite the equation in standard form?
30. What happens if the coefficient of y' is not 1?
31. How do you handle an equation of the form $a(x)y' + b(x)y = c(x)$?
32. Why is dividing by $a(x)$ necessary?
33. What is an initial-value problem in this context?
34. How do you apply an initial condition to determine the constant C ?
35. Solve symbolically:
$$y' + p(x)y = 0.$$
36. What is the general solution to the homogeneous equation?
37. How does the homogeneous solution relate to the general solution?
38. What is the difference between homogeneous and nonhomogeneous equations?
39. What happens when $q(x) = 0$?
40. What happens when $q(x) \neq 0$?
41. Why does the solution involve an integral of $\mu(x)q(x)$?
42. What integration techniques are often required in solving these equations?
43. How does the integrating factor method simplify the problem conceptually?
44. What is the geometric interpretation of multiplying by $\mu(x)$?
45. Why does the method always work for first-order linear equations?
46. What are common mistakes when applying the integrating factor method?
47. How can you verify that a solution is correct?
48. In what applications do first-order linear equations appear?
49. How are they used in modeling motion with air resistance?
50. How are they used in electrical circuits?
51. Why are first-order linear equations important in applied mathematics?

4 Calculus Vol2 - 5. Sequences and Series

<https://openstax.org/books/calculus-volume-2/pages/5-introduction>

4.1 Calculus Vol2 - 5.1 Sequences

<https://openstax.org/books/calculus-volume-2/pages/5-1-sequences>

1. What is an infinite sequence? How is it typically denoted?
2. How can a sequence be interpreted as a function? What is its domain?
3. What is the index of a sequence, and what role does it play?
4. What is an explicit formula for a sequence? Give an example.
5. What is a recursive definition of a sequence? How does it differ from an explicit formula?
6. Given a recursive sequence, how can you compute its first few terms?
7. What does it mean to graph a sequence? How does it differ from graphing a function?
8. What is the limit of a sequence?
9. What does it mean for a sequence a_n to converge to a number L ?
10. Write the formal ϵ - N definition of convergence of a sequence.
11. What does it mean for a sequence to diverge?
12. What are some different ways a sequence can diverge?
13. What does it mean for a sequence to diverge to infinity? To negative infinity?
14. Why does changing a finite number of terms of a sequence not affect its convergence?
15. How can limits of functions be used to determine limits of sequences?
16. State the theorem relating $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{n \rightarrow \infty} a_n$ when $a_n = f(n)$.
17. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n}$ using a function-based argument.
18. What happens to r^n as $n \rightarrow \infty$ when:
 - $0 < r < 1$?
 - $r = 1$?
 - $r > 1$?
19. How can limit laws be applied to sequences?
20. How do you find the limit of a sequence that is a sum or product of simpler sequences?
21. What is a bounded sequence?
22. What does it mean for a sequence to be increasing? Decreasing?
23. What is a monotonic (monotone) sequence?
24. State the Monotone Convergence Theorem.
25. Why is the Monotone Convergence Theorem useful?
26. Give an example of a sequence that is bounded and increasing.
27. Give an example of a sequence that is unbounded.
28. Give an example of a sequence that oscillates and does not converge.
29. How can you determine convergence from the behavior of the terms as $n \rightarrow \infty$?

4.2 Calculus Vol2 - 5.2 Infinite Series

<https://openstax.org/books/calculus-volume-2/pages/5-2-infinite-series>

1. What is an infinite series? How is it related to a sequence?
2. How is an infinite series written using summation notation?
3. What is the difference between a sequence a_n and the series $\sum_{n=1}^{\infty} a_n$?
4. What is a partial sum of a series? How is the k -th partial sum S_k defined?
5. How can an infinite series be defined as a limit of partial sums?
6. What does it mean for an infinite series to converge?
7. What does it mean for an infinite series to diverge?
8. What is the relationship between the convergence of a series and the convergence of its sequence of partial sums?
9. What does it mean for a series to diverge to infinity?
10. How can you determine whether a series converges by examining its partial sums?
11. What is a geometric series? Write its general form.
12. What is the common ratio of a geometric series?
13. Under what condition does a geometric series converge?
14. What is the formula for the sum of a convergent geometric series?
15. What happens if the common ratio r of a geometric series satisfies $|r| \geq 1$?
16. How can you derive the formula for the sum of a geometric series?
17. What is a telescoping series?
18. How do terms cancel in a telescoping series?
19. How can you evaluate the sum of a telescoping series?
20. Why are telescoping series often easier to evaluate than other series?
21. Give an example of a telescoping series and describe its behavior.
22. What is the harmonic series? Write its general form.
23. Does the harmonic series converge or diverge?
24. Why is it not sufficient for $a_n \rightarrow 0$ to guarantee that $\sum a_n$ converges?
25. What is the divergence test (nth-term test for divergence)?
26. If $\lim_{n \rightarrow \infty} a_n \neq 0$, what can you conclude about $\sum a_n$?
27. If $\lim_{n \rightarrow \infty} a_n = 0$, what can you conclude about $\sum a_n$?
28. How are infinite series used in applications such as approximations or modeling?
29. How does the concept of an infinite series extend the idea of a finite sum?
30. How can infinite series be used to represent functions?
31. What is the intuitive meaning of “adding infinitely many terms”?
32. How does the order of terms affect the value of a finite sum? Does this idea carry over to infinite series?

4.3 Calculus Vol2 - 5.3 The Divergence and Integral Tests

<https://openstax.org/books/calculus-volume-2/pages/5-3-the-divergence-and-integral-tests>

1. Why is it often difficult to determine convergence of a series by directly evaluating partial sums?
2. What is the divergence test (nth-term test)?
3. State the divergence test formally.
4. Why does $\sum_{n=1}^{\infty} a_n$ converging imply that $a_n \rightarrow 0$?
5. If $\lim_{n \rightarrow \infty} a_n \neq 0$, what can you conclude about the series $\sum a_n$?
6. If $\lim_{n \rightarrow \infty} a_n = 0$, what can you conclude about the series $\sum a_n$?
7. Why is the divergence test only useful for proving divergence and not convergence?
8. Give an example of a series where $a_n \rightarrow 0$ but the series diverges.
9. How do you apply the divergence test in practice to a given series?
10. What kinds of limits of a_n guarantee divergence (nonzero limit, limit does not exist)?
11. What is the idea behind comparing a series to an improper integral?
12. What is the integral test?
13. What conditions must a function $f(x)$ satisfy to apply the integral test (positivity, continuity, monotonicity)?
14. How are the series $\sum_{n=1}^{\infty} a_n$ and the integral $\int_1^{\infty} f(x), dx$ related in the integral test?
15. State the integral test formally.
16. If the improper integral $\int_1^{\infty} f(x), dx$ converges, what can you conclude about the series?
17. If the improper integral diverges, what can you conclude about the series?
18. How can the integral test be used to prove that the harmonic series diverges?
19. What is a p -series? Write its general form.
20. For what values of p does the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?
21. For what values of p does the p -series diverge?
22. How can the integral test be used to establish the convergence properties of p -series?
23. What is the remainder R_N of a series after N terms?
24. How can the integral test be used to estimate the remainder of a series?
25. State the inequality involving the remainder R_N and integrals.
26. How do upper and lower bounds for R_N arise from the integral test?
27. Why is it useful to estimate the error when approximating a series by partial sums?
28. How can you choose N to ensure the error is less than a desired tolerance?
29. What is the geometric interpretation of the integral test using areas under curves?
30. How does monotonicity of $f(x)$ play a role in bounding the series with integrals?
31. In what situations is the integral test particularly useful compared to other tests?

4.4 Calculus Vol2 - 5.4 Comparison Tests

<https://openstax.org/books/calculus-volume-2/pages/5-4-comparison-tests>

1. What is the main idea behind comparison tests for infinite series?
2. For what type of series (in terms of sign of terms) are comparison tests typically used?
3. Why are geometric series and p -series commonly used in comparison tests?
4. State the Direct Comparison Test for series $\sum a_n$ and $\sum b_n$ with positive terms.
5. If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, what can be concluded about $\sum a_n$?
6. If $0 \leq b_n \leq a_n$ and $\sum b_n$ diverges, what can be concluded about $\sum a_n$?
7. In which cases is the Direct Comparison Test inconclusive?
8. How do you choose an appropriate comparison series b_n for a given series a_n ?
9. Why is it useful to compare a series to a p -series of the form $\sum \frac{1}{n^p}$?
10. What is the behavior of the p -series $\sum \frac{1}{n^p}$ for different values of p ?
11. What is the behavior of a geometric series $\sum ar^n$ depending on the value of r ?
12. State the Limit Comparison Test for two series $\sum a_n$ and $\sum b_n$ with positive terms.
13. What does it mean if
$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c,$$
where $0 < c < \infty$?
14. What can be concluded if the limit in the Limit Comparison Test is 0?
15. What can be concluded if the limit in the Limit Comparison Test is ∞ ?
16. When is the Limit Comparison Test more useful than the Direct Comparison Test?
17. Why does the Limit Comparison Test work even when inequalities between a_n and b_n are hard to establish?
18. What are common functions or expressions you simplify when applying the Limit Comparison Test?
19. How do you simplify rational expressions in n when applying comparison tests?
20. How do logarithmic terms (e.g., $\ln n$) affect comparison with p -series?
21. How do roots (e.g., $\sqrt[n]{n}$) affect comparison with p -series?
22. Give an example of a series that can be compared to $\sum \frac{1}{n^2}$ and explain why.
23. Give an example of a series that can be compared to $\sum \frac{1}{n}$ and explain why.
24. What is the general strategy for proving convergence using comparison tests?
25. What is the general strategy for proving divergence using comparison tests?
26. Why must terms a_n and b_n be positive for these tests?
27. Can comparison tests determine the exact sum of a series? Why or why not?
28. How do comparison tests relate to the Integral Test conceptually?
29. What role do asymptotic behaviors of functions play in the Limit Comparison Test?
30. How can you justify that two sequences have the same “growth rate” for comparison?
31. When comparing $\frac{1}{n^2+1}$ to $\frac{1}{n^2}$, why does the comparison work?

4.5 Calculus Vol2 - 5.5 Alternating Series

<https://openstax.org/books/calculus-volume-2/pages/5-5-alternating-series>

1. What is an alternating series? Write its general form.
2. How can an alternating series be written using the factor $(-1)^n$ or $(-1)^{n+1}$?
3. In the representation $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$, what conditions are imposed on b_n ?
4. State the Alternating Series Test (Leibniz Test).
5. What two conditions must the sequence $\{b_n\}$ satisfy for the Alternating Series Test to apply?
6. Why is the condition $\lim_{n \rightarrow \infty} b_n = 0$ necessary for convergence?
7. Does the Alternating Series Test guarantee absolute convergence or only convergence? Explain.
8. Give an example of a series that converges by the Alternating Series Test.
9. What is meant by absolute convergence?
10. What is meant by conditional convergence?
11. What is the relationship between absolute convergence and convergence of a series?
12. Give an example of a series that converges conditionally but not absolutely.
13. How can you test whether an alternating series converges absolutely?
14. If $\sum |a_n|$ diverges but $\sum a_n$ converges, what type of convergence does the series have?
15. What is the alternating harmonic series? Does it converge absolutely or conditionally?
16. Define the n th partial sum S_N of a series.
17. What is the remainder R_N of a series?
18. For an alternating series satisfying the Alternating Series Test, what inequality bounds the remainder R_N ?
19. Write the inequality relating $|R_N|$ and b_{N+1} .
20. What does the remainder estimate tell us about the error when approximating a sum by S_N ?
21. How can you use the remainder estimate to determine how many terms are needed for a desired accuracy?
22. If you approximate the sum of an alternating series by S_N , what is the maximum possible error?
23. Why do partial sums of an alternating series tend to oscillate around the true sum?
24. Describe the behavior of even and odd partial sums in an alternating series that satisfies the test.
25. How can you estimate the sum of an alternating series to within a given tolerance ε ?
26. For the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, how would you bound the error after N terms?
27. What role does monotonicity ($b_{n+1} \leq b_n$) play in the Alternating Series Test?
28. Can an alternating series converge if the sequence b_n is not decreasing? What does the test say about this?
29. Compare the convergence behavior of $\sum \frac{(-1)^{n+1}}{n}$ and $\sum \frac{1}{n}$.
30. Why is it often difficult to compute the exact sum of an alternating series?
31. What practical advantage does the Alternating Series Estimation Theorem provide?
32. How does the Alternating Series Test differ from comparison or ratio tests?
33. When analyzing a series, why is it useful to check absolute convergence first?
34. What happens if $\lim_{n \rightarrow \infty} a_n \neq 0$ for an alternating series?

4.6 Calculus Vol2 - 5.6 Ratio and Root Tests

<https://openstax.org/books/calculus-volume-2/pages/5-6-ratio-and-root-tests>

1. Why is the condition $\lim_{n \rightarrow \infty} a_n = 0$ not sufficient to guarantee convergence of $\sum a_n$?
2. Give an example of two series where $a_n \rightarrow 0$ in both cases, but one converges and the other diverges.
3. What does it mean for the terms of a series to approach zero “fast enough”?
4. State the Ratio Test for a series $\sum a_n$.
5. Define

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

6. What conclusion can be drawn if $\rho < 1$ in the Ratio Test?
7. What conclusion can be drawn if $\rho > 1$ (or $\rho = \infty$) in the Ratio Test?
8. What happens if $\rho = 1$ in the Ratio Test?
9. What type of convergence does the Ratio Test establish when $\rho < 1$?
10. For what kinds of series is the Ratio Test especially useful?
11. Apply the Ratio Test conceptually: what happens to the ratio $\left| \frac{a_{n+1}}{a_n} \right|$ for factorial-type expressions?
12. State the Root Test for a series $\sum a_n$.
13. Define

$$\rho = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

14. What conclusion can be drawn if $\rho < 1$ in the Root Test?
15. What conclusion can be drawn if $\rho > 1$ in the Root Test?
16. What happens if $\rho = 1$ in the Root Test?
17. Compare the Ratio Test and Root Test: what is similar about their conclusions?
18. In what situations is the Root Test particularly useful?
19. How does the Root Test behave for expressions involving powers like $(b_n)^n$?
20. What is absolute convergence, and how do these tests relate to it?
21. Why do both the Ratio and Root Tests use absolute values?
22. How are these tests useful for power series?
23. When should you consider using the Ratio Test over other tests?
24. When should you consider using the Root Test over other tests?
25. Why are these tests often easier to apply than comparison tests?
26. Can the Ratio or Root Test determine conditional convergence?
27. How do these tests relate to geometric series?

5 Calculus Vol2 - 6. Power Series

<https://openstax.org/books/calculus-volume-2/pages/6-introduction>

5.1 Calculus Vol2 - 6.1 Power Series and Functions

<https://openstax.org/books/calculus-volume-2/pages/6-1-power-series-and-functions>

1. What is the general form of a power series centered at $x = 0$?
2. Write the general form of a power series centered at $x = a$.
3. Explain in your own words why a power series can be thought of as an “infinite polynomial.”
4. Give two examples of power series centered at $x = 0$ with explicit formulas for their coefficients c_n .
5. What does it mean for a power series to be “centered at” a specific point?
6. State the three possible convergence behaviors of a power series centered at $x = a$.
7. Define the *interval of convergence* for a power series.
8. Define the *radius of convergence* of a power series.
9. Why does every power series always converge at its center?
10. Using the geometric series

$$\sum_{n=0}^{\infty} x^n,$$

determine its interval and radius of convergence.

11. How can the Ratio Test be used to help find the radius of convergence of a power series?
12. True/False: If a power series converges for some value x_0 , then it converges for all x with $|x - a| < |x_0 - a|$. Explain your answer.
13. What is the basic idea behind representing a given function $f(x)$ with a power series?
14. Give an example of a function that can be expressed with a power series based on geometric series manipulation.
15. How would you represent the function $\frac{1}{1-x}$ as a power series?
16. How would you represent the function $\frac{1}{1+x^2}$ as a power series?
17. Explain the relationship between power series and functions that are infinitely differentiable near their center.
18. For a power series with finite radius of convergence R , what can be said about convergence at the endpoints $x = a \pm R$?
19. How are the endpoints of the interval of convergence typically tested if the Ratio Test is inconclusive?
20. What implications does the convergence of a power series have for the differentiability and integrability of the represented function on its interval of convergence?

5.2 Calculus Vol2 - 6.2 Properties of Power Series

<https://openstax.org/books/calculus-volume-2/pages/6-2-properties-of-power-series>

1. What is the general result about combining two power series

$$\sum_{n=0}^{\infty} c_n x^n \quad \text{and} \quad \sum_{n=0}^{\infty} d_n x^n$$

that converge on the same interval I ? State the conditions for addition, scalar multiplication, and multiplication by a power of x .

2. Suppose two power series converge to functions $f(x)$ and $g(x)$ on an interval I . What function does the series

$$\sum_{n=0}^{\infty} (c_n + d_n) x^n$$

converge to on I ?

3. For a power series $\sum_{n=0}^{\infty} c_n x^n$ that converges to $f(x)$ on an interval I , what is the power series that converges to $b x^m f(x)$ for a real number b and integer $m \geq 0$?
4. If $\sum_{n=0}^{\infty} c_n x^n$ converges to $f(x)$ on an interval I , what is the new series that represents $f(b x^m)$ when $|b x^m|$ is in the interval I ?
5. What theorem guarantees that a power series can be differentiated and integrated term-by-term on its interval of convergence? What does this theorem say about the resulting series and their convergence?
6. Given a power series

$$f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n,$$

write the term-by-term differentiated series for $f'(x)$ in summation form.

7. Given the same $f(x)$ above, write the term-by-term integrated series representing an antiderivative of $f(x)$.
8. Does term-by-term differentiation or integration change the radius of convergence of a power series? Explain and describe what can happen at the endpoints.
9. Using the known geometric series for $\frac{1}{1-x}$, what is the power series representation of $\frac{1}{(1-x)^2}$ found by term-by-term differentiation? Specify its interval of convergence and endpoint behavior.
10. Find a power series representation for $f(x) = \ln(1+x)$ by integrating the power series for f' and find its interval of convergence.
11. Find a power series representation for $f(x) = \tan^{-1}(x)$ by integrating the power series for f' and find its interval of convergence.

5.3 Calculus Vol2 - 6.3 Taylor and Maclaurin Series

<https://openstax.org/books/calculus-volume-2/pages/6-3-taylor-and-maclaurin-series>

1. What is the general form of the Taylor series for a function $f(x)$ centered at $x = a$? Write it using summation notation.
2. How does the Taylor series simplify when the center $a = 0$? What name is given to this special case?
3. For a function $f(x)$ with derivatives of all orders at $x = a$, write the explicit expression for its n -th Taylor polynomial $p_n(x)$ centered at a .
4. Explain why the coefficients of a power series representation of $f(x)$ at $x = a$ must be given by derivatives of f evaluated at a .
5. What condition must hold for a function's Taylor series to equal the original function $f(x)$ on an interval? Define the remainder $R_n(x)$ in this context.
6. State the Uniqueness of Taylor Series theorem: if a power series converges to $f(x)$ on an interval containing a , what can be said about that series?
7. Define the remainder term $R_n(x)$ for a Taylor polynomial approximation and write the formula for the Lagrange form of the remainder.
8. What does Taylor's Theorem with remainder tell us about the error when approximating $f(x)$ using its n -th Taylor polynomial?
9. Given a specific function $f(x)$, describe the steps to find the Maclaurin polynomial of degree n .
10. Find the Maclaurin series for e^x by writing out several derivatives at 0 and using the definition.
11. Find the Maclaurin series for $\sin x$ and $\cos x$. Explain the pattern in the derivatives that allows you to write the general term.
12. How can the ratio test be used to determine the radius of convergence for a Taylor or Maclaurin series?
13. For a given Taylor series, how would you check whether it actually converges to $f(x)$ (not just converges in value)?
14. Suppose you are given a function $f(x)$ with known derivatives at $x = a$. Write a question asking you to compute the 2nd and 3rd Taylor polynomials explicitly and compare them to the function near a .
15. Create a question that asks you to estimate the error in using the 3rd Taylor polynomial to approximate a function value at a given point using the remainder bound.
16. Why does the remainder $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ matter for Taylor series convergence?
17. Provide the Maclaurin series expansion for $\ln(1 + x)$ and specify its interval of convergence.
18. For a given nonzero center a , how would you find the Taylor series of $\ln(x)$ about $x = a$ and determine its interval of convergence?
19. Write a practice question that asks for the Taylor series of a non-elementary function (e.g., e^{-x^2}) and discuss whether it converges to the function.
20. Formulate a question about how to bound the remainder $R_n(x)$ using information about the maximum value of a derivative on an interval.
21. Write a question that asks you to use Taylor's theorem to prove that the Maclaurin series for e^x converges for all real x .
22. Create a problem asking you to approximate a definite integral using a Taylor or Maclaurin series where the antiderivative is non-elementary.

5.4 Calculus Vol2 - 6.4 Working with Taylor Series

<https://openstax.org/books/calculus-volume-2/pages/6-4-working-with-taylor-series>

1. What is the Taylor series of a function f centered at $x = a$?
2. What is the Maclaurin series? How does it relate to the Taylor series?
3. Write the Taylor series of f centered at a using sigma notation.
4. What is the n th Taylor polynomial $T_n(x)$ of a function f centered at a ?
5. What is the relationship between the Taylor polynomial $T_n(x)$ and the full Taylor series?
6. Compute the Maclaurin series of e^x .
7. Compute the Maclaurin series of $\sin x$.
8. Compute the Maclaurin series of $\cos x$.
9. Compute the Maclaurin series of $\frac{1}{1-x}$ and state its interval of convergence.
10. How can you obtain the Maclaurin series for $\ln(1+x)$ from the geometric series?
11. How can you obtain the Maclaurin series for $\arctan x$?
12. Given a known power series, how can you:
 - (a) Differentiate it term-by-term?
 - (b) Integrate it term-by-term?
13. What happens to the radius of convergence when you differentiate or integrate a power series?
14. How can you construct the Taylor series for e^{2x} using the known series for e^x ?
15. How can you construct the Taylor series for $\sin(3x)$?
16. How can you construct the Taylor series for xe^x ?
17. How can you construct the Taylor series for $\frac{x}{1-x^2}$?
18. How do you find the Taylor series of a function centered at a point $a \neq 0$?
19. Find the Taylor series of $\ln x$ centered at $x = 1$.
20. What substitution allows you to rewrite $\ln x$ in terms of $\ln(1+u)$?
21. How can you approximate a function using its Taylor polynomial?
22. What is Taylor's theorem?
23. What is the remainder term $R_n(x)$ in Taylor's theorem?
24. State the Lagrange form of the remainder.
25. How can you use the remainder term to bound the error of a Taylor approximation?
26. Suppose $|f^{(n+1)}(x)| \leq M$ on an interval containing a and x . What inequality bounds $|R_n(x)|$?
27. How do you determine how large n must be to guarantee a given approximation error?
28. Use a Taylor polynomial to approximate $e^{0.1}$. How can you estimate the error?
29. Use a Taylor polynomial to approximate $\sin(0.2)$. How can you estimate the error?
30. What does it mean for a function to be equal to its Taylor series?
31. Give an example of a function whose Taylor series converges but does not equal the function.

32. Why does the Taylor series of e^x converge for all x ?
33. How does the factorial in the denominator affect convergence of Taylor series?
34. When approximating near $x = a$, why is it better to center the Taylor polynomial at a rather than at 0?

6 Calculus Vol2 - 7. Parametric Equations and Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-introduction>

6.1 Calculus Vol2 - 7.1 Parametric Equations

<https://openstax.org/books/calculus-volume-2/pages/7-1-parametric-equations>

1. What is a parametric equation, and how does it differ from a rectangular equation $y = f(x)$?
2. What is a parameter, and what does it typically represent in applications (e.g., time)?
3. How are x and y expressed in terms of a parameter t in parametric form?
4. How does a point $(x(t), y(t))$ move as the parameter t increases?
5. Why are parametric equations useful for describing curves that are not functions?
6. What does the orientation of a parametric curve mean, and how is it determined?
7. Why can the same curve have multiple different parametric representations?
8. How can you eliminate the parameter to convert parametric equations into a rectangular equation?
9. What is the purpose of eliminating the parameter?
10. How do restrictions on the parameter t affect the portion of the curve that is traced?
11. What are the parametric equations of a circle, and how do they relate to trigonometric functions?
12. How can you verify that a parametric curve represents a known geometric shape?
13. How do parametric equations provide more information than a rectangular equation (e.g., direction of motion)?
14. What does it mean for a curve to be traced once, multiple times, or partially?
15. How is the derivative $\frac{dy}{dx}$ computed for a parametric curve?
16. Why is the formula $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ valid?
17. What happens when $\frac{dx}{dt} = 0$ in the formula for $\frac{dy}{dx}$?
18. How can you find points where the tangent line is horizontal or vertical?
19. What is a critical point for a curve defined parametrically?
20. How can you find the equation of a tangent line to a parametric curve at a given value of t ?
21. How do you compute $\frac{dx}{dt}$ and $\frac{dy}{dt}$ for given parametric equations?
22. What is the relationship between parametric equations and vector-valued functions?
23. How can parametric equations describe curves that cannot be written as $y = f(x)$?

6.2 Calculus Vol2 - 7.2 Calculus of Parametric Curves

<https://openstax.org/books/calculus-volume-2/pages/7-2-calculus-of-parametric-curves>

1. How do you compute the derivative $\frac{dy}{dx}$ for a parametric curve defined by $x(t)$ and $y(t)$?
2. Why is the formula $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ valid, and what assumption is required about $\frac{dx}{dt}$?
3. How do you find the equation of the tangent line to a parametric curve at a specific value $t = t_0$?
4. How can you determine where a parametric curve has a horizontal tangent?
5. How can you determine where a parametric curve has a vertical tangent?
6. What happens to $\frac{dy}{dx}$ when $\frac{dx}{dt} = 0$?
7. How do you compute the second derivative $\frac{d^2y}{dx^2}$ for a parametric curve?
8. Why is the second derivative useful when analyzing concavity of parametric curves?
9. How can you interpret parametric equations as describing motion of a particle?
10. What are the velocity components of a particle moving along a parametric curve?
11. How is speed related to $x'(t)$ and $y'(t)$?
12. How do you compute the arc length of a parametric curve between $t = a$ and $t = b$?
13. What is the formula $L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$ and how is it derived conceptually?
14. How do you compute the area under a parametric curve?
15. Why is the formula $A = \int_{t_1}^{t_2} y(t), x'(t), dt$ used for area?
16. How does the sign of $x'(t)$ affect the computed area?
17. What conditions must hold for the area formula for parametric curves to be valid?
18. How do you handle curves that are traced more than once when computing area?
19. How can you determine whether a parametric curve is traced once or multiple times?
20. How do you compute the surface area generated by revolving a parametric curve about the x -axis?
21. What is the formula $S = 2\pi \int_a^b y(t) \sqrt{(x'(t))^2 + (y'(t))^2} dt$ and when is it used?
22. How does the surface area formula change when revolving around the y -axis?
23. What geometric idea underlies the surface area of revolution for parametric curves?

6.3 Calculus Vol2 - 7.3 Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-3-polar-coordinates>

1. What is the polar coordinate system, and how does it differ from the Cartesian coordinate system?
2. What do the coordinates (r, θ) represent in the polar coordinate system?
3. What is the geometric meaning of the radial coordinate r ?
4. What is the geometric meaning of the angle θ ?
5. What are the pole and the polar axis?
6. How are positive and negative angles interpreted in polar coordinates?
7. What happens when the radial coordinate r is negative?
8. Why does a single point in the plane have infinitely many polar representations?
9. How can the same point be represented using (r, θ) and $(r, \theta + 2\pi)$?
10. How can a point be represented using a negative radius and a shifted angle?
11. What are the conversion formulas from polar to rectangular coordinates?
12. What are the conversion formulas from rectangular to polar coordinates?
13. How can you graph a polar equation $r = f(\theta)$?
14. Why are polar coordinates especially useful for describing circular or radial patterns?
15. What types of curves are naturally expressed in polar coordinates (e.g., circles, spirals)?
16. What does it mean for a polar graph to have symmetry about the pole?
17. What does it mean for a polar graph to have symmetry about the line $\theta = \frac{\pi}{2}$?
18. How can symmetry help simplify graphing polar equations?
19. What is a polar equation, and how does it define a curve?
20. How do you interpret the equation $r = c$ geometrically?
21. How do you interpret the equation $\theta = c$ geometrically?

6.4 Calculus Vol2 - 7.4 Area and Arc Length in Polar Coordinates

<https://openstax.org/books/calculus-volume-2/pages/7-4-area-and-arc-length-in-polar-coordinates>

1. How is the concept of area in polar coordinates different from area in rectangular coordinates?
2. Why are sectors of circles used instead of rectangles in deriving area formulas in polar coordinates?
3. How do you approximate the area of a polar region using a Riemann sum?
4. What is the formula for the area of a region defined by $r = f(\theta)$ on $[\alpha, \beta]$?
5. Why does the factor $\frac{1}{2}$ appear in the polar area formula?
6. What geometric quantity does $(f(\theta))^2$ represent in the area formula?
7. How do you compute the area enclosed by a single polar curve?
8. How do you find the area between two polar curves?
9. Why must you find points of intersection when computing area between two polar curves?
10. How do you determine which curve is the outer curve and which is the inner curve?
11. What happens if a polar curve is traced more than once when computing area?
12. How can symmetry be used to simplify area calculations in polar coordinates?
13. What is the geometric interpretation of the integral $\frac{1}{2} \int_{\alpha}^{\beta} r^2, d\theta$?
14. How is arc length defined for a polar curve $r = f(\theta)$?
15. Why does the arc length formula involve both r and $\frac{dr}{d\theta}$?
16. How can the arc length formula be derived conceptually from parametric equations?
17. What is the relationship between polar coordinates and parametric form when computing arc length?
18. How do you compute $\frac{dr}{d\theta}$ for a given polar function?
19. Why does the expression $\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$ represent speed along the curve?
20. How can you verify that a polar curve encloses a finite area?
21. What are the key similarities and differences between area and arc length formulas in rectangular, parametric, and polar forms?
22. Why are polar coordinates particularly useful for computing areas of radially symmetric regions?
23. What types of curves (e.g., cardioids, roses, spirals) are naturally analyzed using these formulas?

6.5 Calculus Vol2 - 7.5 Conic Sections

<https://openstax.org/books/calculus-volume-2/pages/7-5-conic-sections>

1. What is a conic section, and how is it formed geometrically?
2. What are the four main types of conic sections?
3. How does slicing a cone at different angles produce different conic sections?
4. What is the definition of a conic section in terms of a focus and a directrix?
5. What is a focus, and how is it used to define a conic section?
6. What is a directrix, and what role does it play in defining conics?
7. What is eccentricity, and how is it defined as a ratio of distances?
8. How does the value of eccentricity determine the type of conic (circle, ellipse, parabola, hyperbola)?
9. What is the eccentricity of a circle, ellipse, parabola, and hyperbola?
10. What is the geometric definition of a parabola in terms of focus and directrix?
11. What is the standard equation of a parabola, and what do its parameters represent?
12. How does the parameter p relate to the vertex and focus of a parabola?
13. What is the geometric definition of an ellipse?
14. What is the standard equation of an ellipse centered at (h, k) ?
15. What are the major and minor axes of an ellipse?
16. How are the quantities a , b , and c related in an ellipse?
17. Where are the foci located in an ellipse?
18. What is the geometric definition of a hyperbola?
19. What is the standard equation of a hyperbola centered at (h, k) ?
20. How are the quantities a , b , and c related in a hyperbola?
21. Where are the foci and vertices located in a hyperbola?
22. What are asymptotes, and why are they important for hyperbolas?
23. What is the general second-degree equation of a conic section?
24. What is the discriminant $4AC - B^2$, and how is it used to classify conics?
25. How can you identify a conic section from its general quadratic equation?
26. How can you convert a general conic equation into standard form?
27. What role does completing the square play in rewriting conic equations?
28. How are conic sections represented in polar coordinates?
29. What is the polar form of a conic section involving eccentricity e and parameter p ?
30. How does the equation $r = \frac{ep}{1 \pm e \cos \theta}$ or $r = \frac{ep}{1 \pm e \sin \theta}$ describe conics?
31. What is the focal parameter p , and what is its geometric meaning?
32. How can you determine the type of conic from its polar equation?
33. What are the key differences between conics in rectangular form and polar form?
34. How does changing parameters in the equations affect the shape and position of a conic?

7 Multivariable Calculus

7.1 Basic Types of Functions

1. What is a scalar-valued function of a scalar variable $f : \mathbb{R} \rightarrow \mathbb{R}$?
2. What is a scalar-valued function of several variables? What is a scalar-valued function of a vector variable $f : \mathbb{R}_n \rightarrow \mathbb{R}$?
3. What is a vector-valued function of one variable? What is a vector-valued function of a scalar variable $f : \mathbb{R} \rightarrow \mathbb{R}_n$?
4. What is a vector-valued function of several variables? What is a vector-valued function of a vector variable $f : \mathbb{R}_n \rightarrow \mathbb{R}_m$?
5. What is a scalar field? What is a vector field?
6. When can two functions be composed? State the condition in terms of domains and codomains.
7. Give examples of composing: (a) a scalar-valued function with a vector-valued function, (b) two vector-valued functions.
8. What is the difference between a coordinate transformation and a vector field?

7.2 Vector-Valued Functions of One Variable (Curves)

9. How do we define the limit of a vector-valued function $\mathbf{r}(t)$?
10. How do we define continuity of a vector-valued function?
11. How do we define the derivative of $\mathbf{r}(t)$? Give the definition from first principles.
12. What is the geometric meaning of $\mathbf{r}'(t)$?
13. What is the equation of the tangent line to a curve at t_0 ?
14. State the differentiation rules (sum, scalar multiplication, dot product, cross product).

- (a) What is the derivative of the sum of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) + \mathbf{s}(t)) ?$$

- (b) What is the derivative of a scalar function $f(t)$ times a vector-valued function $\mathbf{r}(t)$?

$$\frac{d}{dt}(f(t) \mathbf{r}(t)) ?$$

- (c) What is the derivative of the dot product of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) \cdot \mathbf{s}(t)) ?$$

- (d) What is the derivative of the cross product of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) \times \mathbf{s}(t)) ?$$

15. Prove that if $|\mathbf{r}(t)| = c$, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.
16. How do we define the integral of a vector-valued function?
17. Define the arc length of a curve.
18. What does it mean to parameterize a curve by arc length?
19. What is the unit tangent vector $\mathbf{T}$?

20. What is the unit normal vector $\mathbf{N}$? Why is it orthogonal to $\mathbf{T}$?
21. What is the unit binormal vector $\mathbf{B}$?
22. What is the osculating plane?
23. What is the normal plane?
24. What is curvature? Give its definitions.
25. What is the curvature of a line? Of a circle?
26. What are position, velocity, acceleration, and speed?
27. Decompose acceleration into tangential and normal components.
28. Give examples of purely tangential and purely normal acceleration.
29. What is torsion? What does it measure?
30. What are the units of curvature and torsion?

7.3 Functions of Several Variables (Scalar Fields)

31. What is a level set of $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
32. What is a level curve? What is a level surface?
33. How can level surfaces be reduced to level curves?
34. How do we define the limit of $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
35. What does it mean that a limit is independent of path?
36. How do we define continuity?
37. What is a partial derivative? Give the definition.
38. What is the geometric interpretation of partial derivatives?
39. What does it mean for a function to be differentiable?
40. What is the total differential?
41. What is the linear approximation of a function?
42. Describe composition with other functions and with a path $\mathbf{r}(t)$.
43. State the chain rule (general and along a path).
44. What are independent and dependent variables in implicit equations?
45. Given $F(x, y) = 0$, derive $\frac{dy}{dx}$ and $\frac{dx}{dy}$.
46. What is the relationship between $\frac{dy}{dx}$ and $\frac{dx}{dy}$?

7.4 Gradients, Geometry, and Optimization

47. What is the gradient ∇f ? Give its type (domain, codomain) and notation.
48. What is the geometric interpretation of the gradient?
49. Why is the gradient orthogonal to level curves? Prove it.
50. Why is the gradient orthogonal to level surfaces? Prove it.
51. What is the directional derivative?
52. How is the directional derivative related to the gradient?
53. In which direction is the directional derivative maximal?
54. What is the derivative of f along a path?
55. What is a critical point?
56. State Fermat's theorem.
57. What is the second derivative test?
58. What is the equation of a tangent line to a level curve?
59. What is the equation of a tangent plane to a level surface?
60. What is the equation of a normal line?
61. What is the equation of the tangent plane to $z = f(x, y)$?

7.5 Implicit Functions and Constrained Optimization

62. What problem does the Implicit Function Theorem solve?
63. State the Implicit Function Theorem for $F(x, y) = 0$.
64. What condition ensures that y can be written as a function of x ?
65. Derive $\frac{dy}{dx} = -\frac{F_x}{F_y}$.
66. What is the geometric meaning of $F_y \neq 0$?
67. What happens if $F_y = 0$?
68. Extend the theorem to $F(x, y, z) = 0$.
69. Express $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
70. How does the theorem justify implicit differentiation?
71. Give examples where the theorem applies and fails.
72. What is the problem of constrained optimization?
73. State the method of Lagrange multipliers.
74. What system of equations must be solved?
75. Why must $\nabla f = \lambda \nabla g$?
76. What is the geometric interpretation of this condition?
77. How does the method extend to higher dimensions?
78. What happens if $\nabla g = \mathbf{0}$?

7.6 Vector-Valued Functions of Several Variables and Jacobians

79. What is a vector-valued function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
80. What is a vector field?
81. What is a parametric surface?
82. How do we compute tangent vectors to a parametric surface?
83. What is the tangent plane to a parametric surface?
84. What is the Jacobian matrix?
85. How is it defined in terms of partial derivatives?
86. What do the rows represent?
87. What do the columns represent?
88. When does the Jacobian reduce to the gradient?
89. What is the total derivative?
90. How does the Jacobian represent linear approximation?
91. What is the Jacobian of the identity map?
92. What is the shape of the Jacobian for $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
93. When is the Jacobian invertible?
94. What is the Jacobian determinant?
95. What is its geometric meaning?
96. How is it used in change of variables?
97. What does it mean when it is zero?
98. State the Inverse Function Theorem (informally).
99. What condition ensures local invertibility?

7.7 Multiple Integrals

100. What is a double integral of a function $f(x, y)$ over a region R ?
101. How is the double integral defined using Riemann sums?
102. What is the geometric interpretation of a double integral?
103. What types of quantities can be computed using double integrals?
104. What is the difference between Type I and Type II regions?
105. How do we compute a double integral as an iterated integral?
106. State Fubini's theorem.
107. When can the order of integration be changed?
108. How is the area of a region computed using double integrals?
109. How is the average value of a function over a region defined?
110. What is a triple integral?
111. What is the geometric interpretation of a triple integral?

112. How can triple integrals be written as iterated integrals?
113. How is volume computed using triple integrals?
114. What are cylindrical coordinates?
115. What are spherical coordinates?
116. How do we convert between Cartesian and cylindrical coordinates?
117. How do we convert between Cartesian and spherical coordinates?
118. What is the Jacobian for cylindrical coordinates?
119. What is the Jacobian for spherical coordinates?
120. What is the change-of-variables formula for multiple integrals?
121. What is the geometric interpretation of the Jacobian determinant in integration?

8 Implicit Differentiation, Gradients, Level Curves, Related Rates

1. What does it mean geometrically for a function to satisfy $F(x, y) = c$?
2. If you move along a curve defined by $F(x, y) = c$, what is $\frac{d}{dx}F(x, y(x))$ and why?
3. How does the chain rule lead to the formula

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0?$$

4. What does the vector $(1, \frac{dy}{dx})$ represent geometrically?
5. What is the gradient ∇F , and what information does it encode?
6. Rewrite implicit differentiation in dot product form. What does

$$\nabla F \cdot (1, \frac{dy}{dx}) = 0$$

mean geometrically?

7. Why must the gradient be perpendicular to level curves?
8. For the circle $F(x, y) = x^2 + y^2$, what is ∇F ? What direction does it point?
9. For the same circle, compute $\frac{dy}{dx}$ using implicit differentiation.
10. Verify explicitly that the gradient is orthogonal to the tangent direction for the circle.
11. If $x = x(t)$ and $y = y(t)$ satisfy $F(x, y) = c$, what is $\frac{dF}{dt}$ and why?
12. Derive the related rates equation

$$\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} = 0.$$

13. What does the vector $(\frac{dx}{dt}, \frac{dy}{dt})$ represent geometrically?
14. Rewrite related rates in dot product form. What does

$$\nabla F \cdot (x', y') = 0$$

mean?

15. What is the relationship between velocity vectors and level curves?
16. How are implicit differentiation and related rates mathematically similar?
17. What is a directional derivative, and how does it relate to $\nabla F \cdot \mathbf{v}$?
18. Why does $\nabla F \cdot \mathbf{v} = 0$ imply motion along a level curve?
19. In one sentence: how are gradients, level curves, implicit differentiation, and related rates connected?
20. Bonus: If a particle moves along a circle, why must its velocity be perpendicular to the radius?

9 Parametric vs Implicit Curves

9.1 Core understanding

1. What is the difference between a parametric representation and an implicit representation of a curve?
2. What does each representation emphasize: generation of points or membership of points?

9.2 Parametric $\rightarrow$ Implicit

3. Given $x = x(t)$ and $y = y(t)$, what does it mean to eliminate the parameter?
4. Can every parametric curve be written in the form $F(x, y) = 0$? What assumptions are needed?
5. What are examples where eliminating the parameter is:
 - easy?
 - difficult or impossible in closed form?
6. Why might the resulting implicit equation not be unique?

9.3 Implicit $\rightarrow$ Parametric

7. What does it mean to parametrize a curve defined by $F(x, y) = 0$?
8. What condition on F guarantees a local parametrization?
9. How does the Implicit Function Theorem justify this?
10. What role does the condition $\nabla F \neq 0$ play?

9.4 Local vs Global behavior

11. What is the difference between a local and a global parametrization?
12. Why can an implicit curve be parametrized locally but not globally?

9.5 Failure modes / limitations

13. Why can't some implicit curves be written as a single function $y = f(x)$?
14. What issues arise from:
 - multiple branches?
 - self-intersections?
 - cusps?
15. Can a curve with multiple disconnected components be described by a single parametrization?

9.6 Conceptual insight

16. Why is it easier to go from parametric to implicit than the other way around?
17. How does an implicit equation define a level set?
18. How do these ideas connect to level curves and gradients?

9.7 Deeper reflection

19. Can you think of a curve that:
 - has a simple parametrization but a complicated implicit form?
 - has a simple implicit form but requires multiple parametrizations?
20. What does this tell you about the trade-offs between representations?

10 Tangent Planes

1. Compare the tangent line in single-variable calculus with the tangent plane in multivariable calculus.
2. What is the definition of a tangent plane to a surface $z = f(x, y)$ at a point (x_0, y_0) ?
3. How do the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ relate to the slope of the tangent plane?
4. Compute the tangent plane to $f(x, y) = x^2 + y^2$ at the point $(1, 2)$.
5. What is the geometric meaning of the linear approximation

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)?$$

6. How is the tangent plane related to the total derivative of $f(x, y)$?
7. Explain how the gradient vector $\nabla f(x_0, y_0)$ is connected to the tangent plane.
8. For a surface given implicitly by $F(x, y, z) = 0$, how do you find the equation of the tangent plane at a point?
9. What role does the gradient $\nabla F(x_0, y_0, z_0)$ play in the implicit surface case?
10. Compute the tangent plane to the surface $x^2 + y^2 + z^2 = 9$ at the point $(1, 2, 2)$.
11. What is the normal vector to the tangent plane of $z = f(x, y)$?
12. How can you express the tangent plane using a normal vector and a point?
13. What is the relationship between the tangent plane and directional derivatives?
14. How can you use the tangent plane to approximate the value of a function near a point?
15. Under what circumstances might a function fail to have a tangent plane at a point?
16. Given a parameterized surface $\mathbf{r}(u, v)$, how do you compute the tangent plane at a point?
17. What vectors span the tangent plane in the parameterized surface case?
18. Compute the tangent plane to the parameterized surface $\mathbf{r}(u, v) = (u, v, u^2 + v^2)$ at $(1, 1, 2)$.
19. How is the cross product used in finding the normal vector to a parameterized surface?
20. What is the difference between a tangent plane and a secant plane?
21. How does the tangent plane behave near critical points where $\nabla f = 0$?
22. Why is the tangent plane considered the best linear approximation of a surface near a point?

11 Directional Derivatives

1. What is the definition of the directional derivative of a function $f(x, y)$ at a point (x_0, y_0) in the direction of a vector $\mathbf{v}$?
2. Why must the direction vector be normalized when computing a directional derivative?
3. Write the limit definition of the directional derivative.
4. How is the directional derivative related to the gradient $\nabla f(x_0, y_0)$?
5. State the formula for the directional derivative in terms of the gradient and a unit vector $\mathbf{u}$.
6. What is the geometric interpretation of the directional derivative?
7. Compute the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 2)$ in the direction of $\mathbf{v} = (3, 4)$.
8. What is the direction of maximum increase of a function at a point?
9. What is the maximum value of the directional derivative at a point, and how is it computed?
10. In what direction is the directional derivative equal to zero?
11. How are level curves of a function related to directional derivatives?
12. Explain why the gradient is perpendicular to level curves.
13. How can directional derivatives be used to test whether a function is constant along a curve?
14. What is the relationship between the directional derivative and the total derivative?
15. How do you compute the directional derivative for a function $f(x, y, z)$ in $\mathbb{R}^3$?
16. Compute the directional derivative of $f(x, y, z) = xyz$ at $(1, 1, 1)$ in the direction of $\mathbf{v} = (1, 1, 1)$.
17. What happens if the gradient at a point is the zero vector?
18. Can a function have all directional derivatives at a point but still fail to be differentiable there?
19. How is the directional derivative related to partial derivatives?
20. Show how the directional derivative can be expressed as a dot product.
21. What is the difference between directional derivatives and partial derivatives?
22. How does the chain rule relate to directional derivatives?
23. Given a parameterized curve $\mathbf{r}(t)$, how do you compute the rate of change of f along the curve?
24. Explain the concept of the derivative of f along a path.
25. How can directional derivatives be interpreted as slopes of tangent lines to curves on surfaces?

12 Total Derivatives

12.1 Type, Domain, and Codomain

1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. What is the type (domain and codomain) of the total derivative $Df(\mathbf{x})$ at a point $\mathbf{x} \in \mathbb{R}^n$?
2. How can you interpret $Df(\mathbf{x})$ as a linear map between tangent spaces?
3. In coordinates, how is $Df(\mathbf{x})$ represented as a matrix? What are its dimensions?

12.2 Jacobian as a Limit (First Principles)

4. Starting from the definition of differentiability, write the limit definition of $Df(\mathbf{x})$.
5. Show that if f is differentiable at $\mathbf{x}$, then there exists a matrix $J_f(\mathbf{x})$ such that

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - J_f(\mathbf{x})\mathbf{h}\|}{\|\mathbf{h}\|} = 0.$$

6. How do the entries of $J_f(\mathbf{x})$ arise from partial derivatives?
7. Why does this limit definition guarantee that the Jacobian gives the *best linear approximation*?

12.3 Directional Derivative and Gradient

8. How does the total derivative differ from a partial derivative?
9. How does the total derivative relate to the gradient ∇f ?
10. What happens to the total derivative if one of the variables is held constant?
11. How is the total derivative related to directional derivatives?
12. Define the directional derivative of f at $\mathbf{x}$ in the direction $\mathbf{v}$ using a limit.
13. Show that the directional derivative can be expressed as

$$Df(\mathbf{x})[\mathbf{v}] = J_f(\mathbf{x})\mathbf{v}.$$

14. In the scalar-valued case $f : \mathbb{R}^n \rightarrow \mathbb{R}$, how is the Jacobian related to the gradient $\nabla f(\mathbf{x})$?
15. Why does the gradient point in the direction of steepest ascent?

12.4 Tangent Planes and Local Linearization

16. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, write the equation of the tangent hyperplane at $\mathbf{x}_0$.
17. How does the linear map $Df(\mathbf{x}_0)$ define this hyperplane?
18. What is the first-order Taylor approximation of f near $\mathbf{x}_0$?
19. Why is $Df(\mathbf{x}_0)$ called the *local linearization* of f ?

12.5 Normal Vector

20. For a level set $f(\mathbf{x}) = c$, what vector is normal to the surface at $\mathbf{x}$?
21. How is this normal vector expressed in terms of the Jacobian?
22. In the case $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, how does ∇f relate to the tangent line?

12.6 Linear Map and Best Linear Approximation

23. What does it mean for $Df(\mathbf{x})$ to be a linear map?
24. What exactly is being mapped by $Df(\mathbf{x})$?
25. Why is $Df(\mathbf{x})$ the *best* linear approximation to f near $\mathbf{x}$?
26. How does the error behave compared to $\|\mathbf{h}\|$?

12.7 Jacobian Determinant and Change of Variables

27. For $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, what is the Jacobian determinant $\det J_f(\mathbf{x})$?
28. How does $\det J_f(\mathbf{x})$ describe local volume (or area) scaling?
29. Why does $|\det J_f(\mathbf{x})|$ appear in the change of variables formula for multiple integrals?
30. What happens geometrically when $\det J_f(\mathbf{x}) = 0$?

12.8 Multivariable Chain Rule

31. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$. State the chain rule for $D(g \circ f)(\mathbf{x})$.
32. Show that

$$J_{g \circ f}(\mathbf{x}) = J_g(f(\mathbf{x})) J_f(\mathbf{x}).$$

33. How does this relate to composition of linear maps?

12.9 Connection to Taylor Series

34. Write the first-order Taylor expansion of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{x}_0$.
35. Where does the Jacobian appear in this expansion?
36. How would higher-order derivatives extend this approximation?

12.10 Newton's Method for Systems

37. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Write the Newton iteration for solving $F(\mathbf{x}) = \mathbf{0}$.
38. How is the Jacobian matrix used in each iteration?
39. Why does Newton's method rely on local linearization?
40. What conditions on $J_F(\mathbf{x})$ are required for the method to work?

12.11 Mapping Tangent Vectors

41. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ be a differentiable curve with $\gamma(0) = x_0$. What is the type (domain and codomain) of $\gamma'(0)$ and of $Df(x_0)\gamma'(0)$?
42. Given a curve $\gamma(t)$ with $\gamma(0) = x_0$, write the expression for the tangent vector to the image curve $f(\gamma(t))$ at $t = 0$. Which theorem justifies this expression?
43. State and explain the identity

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = Df(x_0) \gamma'(0).$$

What are the roles of each term in this equation?

44. Why does the Jacobian $Df(x_0)$ depend only on the tangent vector $\gamma'(0)$ and not on the specific curve γ ? What does this imply about its geometric meaning?

45. Suppose two curves $\gamma_1(t)$ and $\gamma_2(t)$ satisfy $\gamma_1(0) = \gamma_2(0) = x_0$ and $\gamma_1'(0) = \gamma_2'(0)$. What can you say about the derivatives of their image curves under f at $t = 0$?
46. Show that for any vector $v \in \mathbb{R}^n$, the curve $\gamma(t) = x_0 + tv$ satisfies

$$Df(x_0)v = \left. \frac{d}{dt}f(x_0 + tv) \right|_{t=0}.$$

What does this tell you about the relationship between the Jacobian and directional derivatives?

47. In what sense can the Jacobian $Df(x_0)$ be viewed as a map between tangent spaces? Write explicitly the mapping it defines.
48. Explain what is meant by the statement: “the Jacobian maps tangent vectors at x_0 to tangent vectors at $f(x_0)$.” What are the spaces involved?
49. Consider all curves passing through x_0 . What structure do their tangent vectors at $t = 0$ form? How does the Jacobian act on this structure?
50. Why is the Jacobian called the “best linear approximation” of f at x_0 ? How does the curve-based interpretation support this claim?
51. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and let $\gamma(t)$ be a curve through x_0 . Describe geometrically how the tangent vector $\gamma'(0)$ is transformed by $Df(x_0)$.
52. How does the mapping of tangent vectors by $Df(x_0)$ lead to the notion of a tangent line, plane, or hyperplane to the image of f ?
53. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, how does $Df(x_0)$ relate to the gradient $\nabla f(x_0)$? How does this affect the interpretation of $Df(x_0)v$?
54. Explain why the Jacobian provides a linear map that approximates how small displacements near x_0 are transformed under f .

13 The D Notation

1. What is the domain and codomain of the function $f(x, y) = x^2 + y^2$?
2. What is the domain and codomain of the derivative $Df(x, y)$?
3. In general, for a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, what is the type (domain and codomain) of $Df(x)$?
4. Why is $Df(x, y)$ considered a linear map rather than just a collection of partial derivatives?
5. Compute the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = x^2 + y^2$.
6. Write the gradient $\nabla f(x, y)$.
7. Express $Df(x, y)$ as a row vector (Jacobian matrix).
8. Let $h = (h_1, h_2) \in \mathbb{R}^2$. Compute $Df(x, y)(h)$.
9. How does the expression $Df(x, y)(h)$ relate to the dot product with the gradient?
10. Write the first-order linear approximation of $f(x, y)$ at a point (x, y) using $Df(x, y)$.
11. What is the geometric meaning of the vector $\nabla f(x, y)$?
12. What does $Df(x, y)(h)$ represent geometrically?
13. How does the definition of $Df(x, y)(h)$ relate to the directional derivative?
14. Compare the expressions:

$$Df(x)(h) = 2x \cdot h \quad \text{and} \quad Df(x, y)(h) = 2x \cdot h_1 + 2y \cdot h_2.$$

What structural similarity do you observe?

15. In the scalar-to-scalar case $f : \mathbb{R} \rightarrow \mathbb{R}$, why can $Df(x)$ be identified with multiplication by a number?
16. Explain why $Df(x, y)$ can be viewed as the “best linear approximation” of f near (x, y) .
17. If $h = (1, 0)$, what does $Df(x, y)(h)$ compute? What about $h = (0, 1)$?
18. How would you interpret $Df(x, y)$ acting on an arbitrary direction vector h in terms of rates of change?

Scalar $\rightarrow$ Scalar	$f : \mathbb{R} \rightarrow \mathbb{R}$	$Df(x)(h) = f'(x) h$
Scalar $\rightarrow$ Vector	$f : \mathbb{R} \rightarrow \mathbb{R}^n$	$Df(x)(h) = f'(x) \cdot h$
Vector $\rightarrow$ Scalar	$f : \mathbb{R}^m \rightarrow \mathbb{R}$	$Df(x)(h) = \nabla f(x) \cdot h$
Vector $\rightarrow$ Vector	$f : \mathbb{R}^m \rightarrow \mathbb{R}^n$	$Df(x)(h) = J_f(x) h$

A hint: Think of D as a type-lifting operator that transforms a function into a new function that returns linear maps

$$D : (\mathbb{R}^m \rightarrow \mathbb{R}^n) \rightarrow (\mathbb{R}^m \rightarrow \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$$

where $\mathcal{L}(\cdot) =$ space of linear maps.

A hint: Think of $Df(x)$ as: “Give me a direction h , and I’ll tell you how f changes in that direction.”

$$Df(x)(h) \approx f(x + h) - f(x)$$

14 Implicit Function Theorem

1. What does it mean for a function to be defined implicitly?
2. How does the implicit function theorem explain why many curves locally look like graphs of functions?
3. Give an example of an equation that defines a relationship between variables implicitly.
4. Why can some implicit equations not be solved algebraically for one variable?
5. What question does the implicit function theorem answer about such equations?
6. State the implicit function theorem for a function $F(x, y)$.
7. What conditions must the function $F(x, y)$ satisfy for the theorem to apply?
8. Why is the condition $F_y(x_0, y_0) \neq 0$ important?
9. What does the theorem guarantee about the existence of a function $y = f(x)$?
10. In what sense does the resulting function exist (global or local)?
11. What does “locally” mean in the statement of the implicit function theorem?
12. How does the theorem justify the use of implicit differentiation?
13. Derive the formula for $\frac{dy}{dx}$ from the equation $F(x, y) = 0$.
14. What geometric feature often occurs when $F_y = 0$?
15. At which points of the circle $x^2 + y^2 = 1$ does the implicit function theorem fail to produce $y = f(x)$?
16. How can the theorem be used to justify solving equations numerically?
17. How does the theorem generalize to functions $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$?
18. What role does the Jacobian matrix play in the multivariable version of the theorem?
19. What rank condition must the Jacobian satisfy?
20. What does the theorem guarantee about expressing some variables as functions of others?
21. How many variables can be solved for in terms of the remaining variables?
22. How does the implicit function theorem relate to the inverse function theorem?
23. In what sense can the inverse function theorem be viewed as a special case of the implicit function theorem?
24. How is the implicit function theorem used in constrained optimization?
25. Why is the theorem important for the method of Lagrange multipliers?
26. How does the theorem help describe surfaces defined by equations $F(x, y, z) = 0$?
27. What geometric object is typically formed by the solution set of such an equation?
28. What does the theorem imply about the smoothness of these solution sets?
29. What would go wrong in calculus if the implicit function theorem were not true?

15 Change of Variables in Multiple Integrals

15.1 Conceptual Foundations

1. What is the general idea behind a change of variables in a double integral?
2. What is meant by a transformation

$$T(u, v) = (x(u, v), y(u, v))?$$

3. What conditions must a transformation $T(u, v)$ satisfy in order for the change of variables theorem to apply?
4. What is the Jacobian matrix of a transformation $T(u, v) = (x(u, v), y(u, v))$?
5. Define the Jacobian determinant:

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)}.$$

6. Write the explicit formula for the Jacobian determinant in terms of partial derivatives.
7. Why does the absolute value of the Jacobian determinant appear in the change-of-variables formula?
8. What geometric meaning does the Jacobian determinant have?
9. What does it mean if the Jacobian determinant is zero at a point?
10. What happens to area elements under a transformation?

15.2 Change of Variables Theorem (Double Integrals)

11. State the Change of Variables Theorem for double integrals.
12. Complete the formula:

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

13. What is the relationship between the regions R and S ?
14. Why is it often easier to integrate over S rather than R ?
15. Describe the general steps required to compute a double integral using change of variables.

15.3 Computing Jacobians

16. Compute the Jacobian determinant of the transformation:

$$x = u^2 - v^2, \quad y = 2uv.$$

17. Compute the Jacobian determinant for:

$$x = au + bv, \quad y = cu + dv.$$

18. For what condition on a, b, c, d is the transformation invertible?
19. How does the Jacobian determinant behave under composition of transformations?
20. If T has Jacobian J_T and its inverse has Jacobian $J_{T^{-1}}$, what is the relationship between them?

15.4 Polar Coordinates

21. Write the transformation from polar to Cartesian coordinates.
22. Compute the Jacobian determinant for the polar transformation:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

23. Why does the area element become

$$dA = r \, dr \, d\theta$$

in polar coordinates?

24. Rewrite the integral

$$\iint_R f(x, y) \, dA$$

in polar coordinates.

25. Describe a region in the plane that is particularly well suited for polar coordinates.

26. Evaluate conceptually:

$$\iint_R (x^2 + y^2) \, dA$$

over a disk using polar coordinates.

15.5 Geometric Interpretation

27. How does a small rectangle in the uv -plane transform under T ?
28. How does the Jacobian determinant approximate local area scaling?
29. What is the geometric interpretation of a negative Jacobian determinant?
30. Why do we take the absolute value in the change of variables formula?

15.6 Strategy and Problem Solving

31. When facing a difficult region R , how do you decide on a good transformation?
32. Why are linear transformations often useful for parallelogram-shaped regions?
33. How can you transform an elliptical region into a circular region?
34. What is the advantage of transforming complicated boundaries into rectangular ones?
35. What common algebra mistakes occur when computing Jacobians?
36. Why is it important to rewrite the integrand entirely in terms of u and v ?
37. What must always be changed: the integrand, the differential, the region?
38. How can you check whether your transformation is one-to-one?

15.7 Extension to Triple Integrals

39. State the change of variables formula for triple integrals.
40. Write the definition of the Jacobian determinant for a transformation

$$(u, v, w) \mapsto (x, y, z).$$

41. How does volume scale under a transformation in $\mathbb{R}^3$?
42. What is the Jacobian determinant for spherical coordinates?
43. What is the Jacobian determinant for cylindrical coordinates?

15.8 Deep Understanding / Proof-Level Questions

44. How can the change of variables theorem be justified using Riemann sums?
45. Why does the determinant appear naturally in linear approximations?
46. How does the change-of-variables formula relate to linear algebra?
47. In what sense is the Jacobian determinant a multidimensional generalization of dx/du ?
48. Why is differentiability essential in the theorem?
49. What would fail if the transformation were not invertible?
50. How does this theorem connect to the substitution rule in single-variable calculus?

16 Jacobian Matrix and Determinant

16.1 Basic Definitions

1. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be given by

$$F(x_1, \dots, x_n) = (f_1(x), \dots, f_m(x)).$$

What is the definition of the *Jacobian matrix* $J_F(x)$?

2. Write explicitly the Jacobian matrix:

$$J_F(x) = \left(\frac{\partial f_i}{\partial x_j} \right).$$

What are its dimensions?

3. In which case is the Jacobian matrix square?
4. When the Jacobian matrix is simply the transpose of the gradient vector?
5. When $m = n$, how is the *Jacobian determinant* defined?
6. What does the notation

$$\det J_F(x)$$

represent?

16.2 Jacobian as Derivative

7. How is the Jacobian matrix related to the total derivative (Fréchet derivative) of a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
8. State the linear approximation formula:

$$F(x+h) \approx F(x) + J_F(x)h.$$

9. What does the Jacobian matrix represent geometrically at a point?
10. What condition on $J_F(x)$ ensures that F is locally invertible near x ?

16.3 Chain Rule

11. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^m \rightarrow \mathbb{R}^p$. State the multivariable chain rule for Jacobians:

$$J_{G \circ F}(x) = ?$$

12. When $n = m = p$, what is the corresponding formula for determinants?
13. Why does this imply

$$\det J_{G \circ F}(x) = \det J_G(F(x)) \cdot \det J_F(x)?$$

16.4 Geometric Interpretation

14. How does the Jacobian determinant describe local volume scaling?
15. What does it mean if $\det J_F(x) > 0$?
16. What does it mean if $\det J_F(x) < 0$?
17. What does it mean if $\det J_F(x) = 0$?
18. How does the absolute value $|\det J_F(x)|$ relate to infinitesimal volume elements?

16.5 Change of Variables in Integration

19. State the change-of-variables formula in $\mathbb{R}^n$:

$$\int_{F(U)} g(y) dy = \int_U g(F(x)) |\det J_F(x)| dx.$$

20. Why is the absolute value necessary?
21. What conditions must F satisfy for the change-of-variables formula to hold?
22. Compute the Jacobian determinant for the transformation from Cartesian to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

23. Why does the area element become

$$dx dy = r dr d\theta?$$

16.6 Special Cases and Examples

24. Compute the Jacobian matrix of

$$F(x, y) = (x^2 y, e^{xy}).$$

25. Compute the Jacobian determinant of

$$F(x, y) = (u, v)$$

where

$$u = x + y, \quad v = x - y.$$

26. For a linear transformation $F(x) = Ax$, what is $J_F(x)$?
27. How is $\det J_F(x)$ related to $\det A$ in the linear case?
28. If F is a rotation in $\mathbb{R}^2$, what is its Jacobian determinant?

16.7 Inverse Function Theorem

29. State the inverse function theorem in terms of the Jacobian determinant.
30. Why does $\det J_F(x_0) \neq 0$ imply local invertibility?
31. How is the Jacobian of the inverse function expressed in terms of J_F ?

16.8 Higher Concepts

32. How does the Jacobian generalize the derivative of a single-variable function?
33. In what sense is the Jacobian matrix the best linear approximation?
34. How does the determinant relate to orientation?
35. What is the Jacobian in the context of manifolds and coordinate changes?
36. How does the Jacobian relate to differential forms?

17 Hessian Matrix

17.1 Basic Definitions

1. What is the Hessian matrix of a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
2. Write the general form of the Hessian matrix for a function $f(x_1, x_2, \dots, x_n)$.
3. What are the entries of the Hessian matrix in terms of partial derivatives?
4. For $f(x, y)$, write the Hessian explicitly.

17.2 Computation

5. Compute the Hessian of $f(x, y) = x^2 + y^2$.
6. Compute the Hessian of $f(x, y) = x^2y + y^3$.
7. Compute the Hessian of $f(x, y, z) = xyz$.
8. Compute the Hessian of $f(x, y) = e^{x^2+y^2}$.

17.3 Interpretation

9. What information does the Hessian matrix encode about a function?
10. How is the Hessian related to the second-order Taylor expansion?
11. What does the Hessian tell you about local curvature?

17.4 Critical Points and Optimization

12. What is a critical point of a multivariable function?
13. State the second derivative test using the Hessian for functions of two variables.
14. What are the conditions on the Hessian for:
 - a local minimum?
 - a local maximum?
 - a saddle point?
15. What does it mean for the Hessian to be positive definite?
16. What does it mean for the Hessian to be negative definite?
17. What happens if the Hessian is indefinite?
18. What happens if the determinant of the Hessian is zero?

17.5 Matrix Properties

19. When is the Hessian symmetric?
20. State the conditions under which mixed partial derivatives are equal.
21. Why is symmetry of the Hessian important?

17.6 Eigenvalues and Geometry

22. How are eigenvalues of the Hessian related to curvature?
23. What does it mean if all eigenvalues are positive?
24. What does it mean if eigenvalues have mixed signs?
25. How do eigenvectors of the Hessian relate to principal directions?

17.7 Advanced Connections

26. How does the Hessian relate to convexity of a function?
27. State the condition for a function to be convex using the Hessian.
28. What is the Hessian in the context of optimization algorithms (e.g., Newton's method)?
29. How is the Hessian used in quadratic approximation?
30. What is the bordered Hessian and when is it used?

17.8 Conceptual Checks

31. Why does the first derivative test fail for multivariable functions without second-order information?
32. Can a function have a zero gradient but not be an extremum? Explain using the Hessian.
33. How does the Hessian generalize the second derivative from single-variable calculus?

18 Frenet–Serret Frame and Formulas

1. What is a regular curve? What condition must a parametrized curve $\mathbf{r}(t)$ satisfy?
2. What is arc-length parametrization? How do we define the arc-length parameter s ?
3. Why is arc-length parametrization especially convenient when defining the Frenet frame?
4. How is the unit tangent vector $\mathbf{T}(s)$ defined in terms of $\mathbf{r}(s)$?
5. Why does $\|\mathbf{T}(s)\| = 1$ always hold?
6. Show that $\mathbf{T}'(s)$ is orthogonal to $\mathbf{T}(s)$. Why must this be true?
7. How is curvature $\kappa(s)$ defined in terms of $\mathbf{T}'(s)$?
8. What is the geometric meaning of curvature? What does it measure intuitively?
9. Under what condition is curvature zero? What does this imply about the curve?
10. How is the principal normal vector $\mathbf{N}(s)$ defined?
11. Why is $\mathbf{N}(s)$ a unit vector?
12. What is the binormal vector $\mathbf{B}(s)$ and how is it defined?
13. Why is $\mathbf{B}(s)$ orthogonal to both $\mathbf{T}(s)$ and $\mathbf{N}(s)$?
14. Show that $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ forms an orthonormal basis.
15. What is the geometric interpretation of the Frenet frame at a point on the curve?
16. Write down the Frenet–Serret formulas.
17. What does the equation $\mathbf{T}'(s) = \kappa(s)\mathbf{N}(s)$ tell us geometrically?
18. How is torsion $\tau(s)$ defined?
19. What is the geometric meaning of torsion?
20. What does it mean if torsion is zero everywhere?
21. Explain the equation $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s)$.
22. Why must $\mathbf{N}'(s)$ involve both $\mathbf{T}(s)$ and $\mathbf{B}(s)$?
23. Derive the Frenet–Serret formulas starting from orthonormality constraints.
24. How do curvature and torsion uniquely determine a curve (up to rigid motion)?
25. What is the fundamental theorem of space curves?
26. Compute κ and τ for a straight line.
27. Compute κ and τ for a circle of radius R .
28. Compute κ and τ for a helix.
29. How do you compute curvature for a curve given in a non arc-length parameter t ?
30. What is the formula for curvature using $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$?
31. How is torsion computed using $\mathbf{r}'(t)$, $\mathbf{r}''(t)$, and $\mathbf{r}'''(t)$?
32. Why does the cross product $\mathbf{r}'(t) \times \mathbf{r}''(t)$ appear in curvature and torsion formulas?
33. What happens to the Frenet frame at points where $\kappa = 0$?
34. Can the Frenet frame always be defined globally along a curve?
35. What is the osculating plane? Which vectors span it?

36. What is the normal plane? Which vectors span it?
37. What is the rectifying plane? Which vectors span it?
38. How does the Frenet frame describe the local geometry of a curve?
39. In what sense does curvature measure bending and torsion measure twisting?
40. How would you reconstruct a curve from given curvature and torsion functions?
41. What role do initial conditions play in reconstructing a curve?
42. How does the Frenet frame evolve along the curve?
43. What is the matrix form of the Frenet–Serret equations?
44. Why is the Frenet–Serret matrix skew-symmetric (up to signs)?
45. How is the Frenet frame related to rigid body motion along a curve?
46. What is the connection between angular velocity and the Frenet–Serret formulas?
47. How can the Frenet frame be interpreted as a moving coordinate system?
48. What breaks down if the curve is not twice differentiable?

19 Geometry of Differential Equations

Let's develop a geometric understanding of differential equations by connecting them to vector fields and tangent structures. Instead of viewing a differential equation purely as an algebraic object, we interpret it as a rule that assigns a direction (velocity) to every point in space. This perspective reveals that solutions to differential equations are not arbitrary functions, but special curves—called integral curves—that follow these directions at every point.

We begin by relating slope fields and vector fields, showing that a first-order differential equation naturally defines a direction field. From there, we introduce the concept of integral curves as trajectories that are everywhere tangent to the vector field. This provides a unifying viewpoint: solving a differential equation means finding curves whose tangent vectors agree with a prescribed field.

We then connect these ideas to differential geometry by interpreting vector fields as tangent vector fields and surfaces as collections of tangent planes. This highlights the distinction between local information (given by the vector field) and global behavior (the resulting trajectories).

Finally, we interpret the Forward Euler method geometrically as a procedure that approximates integral curves by taking small steps in the direction of the vector field. This reveals numerical methods as discrete versions of continuous geometric motion.

Overall, the goal is to build an intuitive and structural understanding of differential equations as geometry: vector fields define motion, and solutions arise by following that motion.

19.1 Vector Fields and Differential Equations

1. What does a vector field $\mathbf{V}(\mathbf{x})$ assign to each point in space?
2. In what sense does a vector field define a “local rule of motion”?
3. How can a differential equation $\frac{d\mathbf{x}}{dt} = \mathbf{V}(\mathbf{x})$ be interpreted geometrically?
4. Why does a vector field determine velocity but not a global destination?
5. How does an initial condition $\mathbf{x}(0) = \mathbf{x}_0$ select a unique trajectory?
6. What does it mean for the family of all solutions to be “encoded” in the vector field?

19.2 Slope Fields and Direction Fields

7. How does the equation $\frac{dy}{dx} = f(x, y)$ define a slope field?
8. How can a slope field be converted into a vector field using $(1, f(x, y))$?
9. Why is a slope field really a direction field rather than a unique vector field?
10. How are solution curves related to the directions shown in a slope field?
11. Why can we scale vectors (e.g. $(\lambda, \lambda f)$) without changing the solutions?

19.3 Integral Curves

12. What is the definition of an integral curve in terms of $\gamma'(t) = \mathbf{V}(\gamma(t))$?
13. Why is an integral curve the same thing as a solution to a differential equation?
14. What does it mean geometrically that a curve “follows” a vector field?
15. Why are integral curves also called trajectories or flow lines?
16. How can an integral curve be written in integral form:

$$\gamma(t) = \gamma(0) + \int_0^t \mathbf{V}(\gamma(s)) ds?$$

17. Why is the term “integral” used in “integral curve”?

19.4 Tangent Vectors and Tangent Vector Fields

18. What is a tangent vector to a curve?
19. What is the tangent plane $T_p S$ at a point on a surface?
20. Why does a tangent plane contain infinitely many possible tangent directions?
21. What is a tangent vector field on a surface S ?
22. Why does a tangent vector field require choosing one vector from each tangent plane?
23. How do parametrizations $\mathbf{r}(u, v)$ produce tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$?
24. How can a general tangent vector be expressed as a linear combination $a\mathbf{r}_u + b\mathbf{r}_v$?
25. How can a vector field in $\mathbb{R}^3$ be projected onto a tangent plane?
26. How do curves on a surface naturally generate tangent vector fields?

19.5 Connection Between Tangent Fields and Differential Equations

27. How does the condition $\dot{\gamma}(t) = \mathbf{V}(\gamma(t))$ relate tangent vectors to vector fields?
28. Why can solution curves be viewed as curves whose tangent vectors match a vector field?
29. In what sense is a differential equation prescribing a tangent vector field?
30. How does this interpretation unify slope fields, vector fields, and tangent vectors?

19.6 Forward Euler Method

31. What is the Forward Euler update rule:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{V}(\mathbf{x}_n)?$$

32. How can this be interpreted as “taking a small step in the direction of the vector field”?
33. Why does Forward Euler approximate integral curves with piecewise linear segments?
34. What assumption does Euler make about the vector field over a small time step?
35. How does the exact solution use the integral:

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \int_t^{t+h} \mathbf{V}(\mathbf{x}(s)) ds?$$

36. Why is Forward Euler equivalent to a left-endpoint Riemann sum?
37. Why does error accumulate in Forward Euler when the vector field changes along the path?

19.7 Local vs Global Behavior

38. What is the difference between a local rule (vector field) and a global object (solution curve)?
39. Why does repeatedly applying local information produce a global trajectory?
40. How does this idea relate to numerical methods like Forward Euler?

19.8 Flows and Deeper Geometric View

41. What is a flow $\varphi_t(\mathbf{x}_0)$ generated by a vector field?
42. How does a vector field define motion over time through its integral curves?
43. How does Forward Euler approximate a continuous flow with a discrete map?
44. Why can differential equations be viewed as describing motion on geometric spaces?
45. How does restricting motion to a surface require the vector field to lie in the tangent plane?

20 Riemann and Darboux Integration

20.1 Foundational Definitions

1. What is a partition of an interval $[a, b]$?
2. Define the norm (mesh size) of a partition.
3. What is a tagged partition?
4. Define a Riemann sum.
5. State the definition of the Riemann integral using Riemann sums.
6. Define the upper Darboux sum.
7. Define the lower Darboux sum.
8. What are the quantities

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

and

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)?$$

9. Define the upper Darboux integral.
10. Define the lower Darboux integral.
11. State the definition of Darboux integrability.
12. Why must a function be bounded to be Riemann integrable?
13. Why must a function be bounded to be Darboux integrable?

20.2 Conceptual Understanding

14. What is the geometric intuition behind a Riemann sum?
15. What is the geometric intuition behind upper and lower Darboux sums?
16. Why does refining a partition generally improve approximation quality?
17. What happens to lower Darboux sums when a partition is refined?
18. What happens to upper Darboux sums when a partition is refined?
19. Why is

$$L(f, P) \leq U(f, P)$$

always true?

20. Explain why

$$\sup_P L(f, P) \leq \inf_P U(f, P).$$

21. What does it mean geometrically when the upper and lower integrals are equal?
22. Why are Darboux sums often considered technically simpler than Riemann sums?
23. Why are Riemann sums closer to the historical intuition of integration?

20.3 Comparisons Between Riemann and Darboux Integration

24. What is the main difference between a Riemann sum and a Darboux sum?
25. In a Riemann sum, how are sample points chosen?
26. In a Darboux sum, why are sample points unnecessary?
27. Which theory uses arbitrary sample points explicitly?
28. Which theory uses suprema and infima on subintervals?
29. What role does the partition norm play in Riemann integration?
30. What role does partition refinement play in Darboux integration?
31. State the theorem relating Riemann integrability and Darboux integrability.
32. Why are the two theories equivalent for bounded functions on closed intervals?
33. Can two different tagged partitions with the same underlying partition produce different Riemann sums?
34. Can two different Darboux sums exist for the same partition?
35. How do upper and lower Darboux sums bound all possible Riemann sums?
36. Explain how Darboux integration avoids explicit limits of sums.
37. Explain how Riemann integration encodes convergence directly through limits.

20.4 Theorems and Properties

38. State the linearity property of the integral.
39. State the additivity over intervals property.
40. State the monotonicity property of the integral.
41. If f is continuous on $[a, b]$, why is it integrable?
42. Why is every monotone function on $[a, b]$ integrable?
43. State a criterion for integrability involving discontinuities.
44. What is the relationship between measure-zero discontinuity sets and integrability?
45. Why is the Dirichlet function not Riemann integrable?
46. Why is the characteristic function of the rationals discontinuous everywhere?
47. Give an example of a function that is discontinuous but still Riemann integrable.
48. Why is a function with finitely many discontinuities integrable?
49. What happens to the integral if two functions differ only on finitely many points?
50. What happens if two functions differ only on a measure-zero set?

20.5 Technical Manipulation

51. Compute the lower and upper Darboux sums for a constant function.
52. Compute the lower and upper Darboux sums for $f(x) = x$ on $[0, 1]$ using a uniform partition.
53. Derive the Riemann sum for $f(x) = x^2$ on $[0, 1]$.
54. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Riemann sums.

55. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Darboux sums.

56. For a uniform partition, how do you compute

$$\Delta x?$$

57. Why do continuous functions attain maxima and minima on closed intervals?
58. Why is compactness important in Darboux integration?
59. What is the difference between taking a supremum over all lower sums and taking a limit?
60. Why does the existence of arbitrarily close upper and lower sums imply integrability?

20.6 Interpreting the Integral

61. What does the expression

$$\sum_i f(x_i) \Delta x_i$$

represent before assigning any geometric or physical interpretation to it?

62. Why is the integral fundamentally an accumulation operator rather than specifically an area operator?
63. In the area interpretation of the integral, what do the quantities

$$f(x_i), \quad \Delta x_i, \quad f(x_i) \Delta x_i$$

represent geometrically?

64. Why does

$$f(x_i) \Delta x_i$$

approximate the area of a thin rectangle?

65. What happens to the Riemann sum as

$$\max \Delta x_i \rightarrow 0?$$

66. Why does the definite integral emerge as a limit of Riemann sums?
67. If

$$g'(x) = f(x),$$

why does

$$f(x) \Delta x$$

approximate the small change

$$\Delta g?$$

68. What is the conceptual meaning of

$$dg = g'(x) dx?$$

69. Why can the same expression

$$f(x)\Delta x$$

be interpreted both as rectangle area and as accumulated change?

70. What determines the interpretation of the integrand $f(x)$?

71. Why is the equality

$$\int_a^b g'(x) dx = g(b) - g(a)$$

an accumulation statement?

72. In the Fundamental Theorem of Calculus, what exactly is being accumulated?

73. Why does integrating a derivative recover net change?

74. How does the telescoping intuition appear in

$$\sum \Delta g_i?$$

75. Define

$$A(x) = \int_a^x f(t) dt.$$

What does $A(x)$ represent geometrically?

76. Why does increasing x by a small amount h add approximately

$$f(x)h$$

to the accumulated area?

77. Use the difference quotient to explain why

$$A'(x) = f(x).$$

78. Why does the identity

$$A'(x) = f(x)$$

connect the area interpretation to the antiderivative interpretation?

79. Why is the Fundamental Theorem of Calculus fundamentally a statement that differentiation and integration are inverse operations?

80. How does the notion of “local rate” connect to “global accumulation”?

81. What is the difference between:

- a function value,
- a derivative,
- a differential,
- an accumulated quantity?

82. Why is

$$\int_a^b f(x) dx$$

not intrinsically an area?

83. Give examples where the integral represents:

- distance,

- mass,
- charge,
- probability,
- energy.

84. In each example above, what plays the role of:

- density/rate,
- small measure,
- accumulated total?

85. Explain the pattern:

$$\text{density} \times \text{small measure} \rightarrow \text{accumulated quantity}.$$

86. Why is the rectangle picture only one geometric realization of integration?

87. What is the conceptual difference between:

$$f(x) \quad \text{and} \quad f(x) dx?$$

88. Why is

$$dx$$

best understood as “a tiny increment of the independent variable”?

89. Why is the statement

$$dg = g'(x) dx$$

more fundamental than the rectangle interpretation?

90. What does the Fundamental Theorem of Calculus reveal about the relationship between local structure and global structure?

20.7 Integrals and Euler’s Method

91. Consider the initial value problem

$$y'(x) = f(x, y(x)), \quad y(a) = y_0.$$

Write down the forward Euler update on a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

and explain why one step has the form

$$y_{i+1} - y_i \approx f(x_i, y_i) \Delta x_i.$$

92. Why is a single Euler update naturally interpreted as an accumulation of a local rate over a small interval?

93. Explain why it is reasonable to call

$$y_{i+1} = y_i + f(x_i, y_i) \Delta x_i$$

an integration step rather than merely an algebraic update.

94. Sum the Euler increments over many steps and derive

$$y_n = y_0 + \sum_{i=0}^{n-1} f(x_i, y_i) \Delta x_i.$$

What is the conceptual meaning of this formula, and how does the initial value y_0 enter as the starting accumulated amount?

95. Why does the exact solution of

$$y'(x) = f(x, y(x)), \quad y(a) = y_0,$$

satisfy the integral equation

$$y(x) = y_0 + \int_a^x f(t, y(t)) dt?$$

How does Euler's method mimic this identity?

96. Why is the solution curve of an ODE often called an integral curve? What is being integrated, and in what sense is the curve determined by that integration?

97. For a general first-order ODE

$$y' = f(x, y),$$

what information does the slope field encode? Why does prescribing an initial value

$$y(a) = y_0$$

select one particular curve from that field?

98. In the autonomous case

$$y' = F(y),$$

how does the interpretation of an integral curve differ from the case

$$y' = f(x, y)?$$

What remains the same?

99. Show that an ordinary definite integral is the special case of the ODE

$$y'(x) = g(x), \quad y(a) = y_0,$$

where the right-hand side does not depend on y . What is the exact solution?

100. Let G be one antiderivative of g , so that

$$G'(x) = g(x).$$

Why are all solutions of

$$y'(x) = g(x)$$

of the form

$$y(x) = G(x) + C?$$

How does the initial condition

$$y(a) = y_0$$

determine the value of C ?

101. Compare the slope field of a general ODE

$$y' = f(x, y)$$

with the slope field of

$$y' = g(x).$$

Why is the latter vertically uniform, meaning that at a fixed x the slope is the same at every height y ?

102. Why does the vertical uniformity of the slope field for

$$y' = g(x)$$

imply that all integral curves are vertical translates of one another? How does this connect to the additive constant C in

$$y = G(x) + C?$$

103. Apply Euler's method to

$$y'(x) = g(x), \quad y(a) = y_0.$$

Why does the scheme become

$$y_{i+1} = y_i + g(x_i) \Delta x_i?$$

104. Iterate the previous update and show that

$$y_n = y_0 + \sum_{i=0}^{n-1} g(x_i) \Delta x_i.$$

Why is this now literally a Riemann sum plus the initial value, rather than only an Euler approximation?

105. What special feature makes the discrete accumulation for

$$y' = g(x)$$

telescope exactly into

$$y_n - y_0 = \sum_{i=0}^{n-1} (y_{i+1} - y_i)?$$

Why does the same telescoping identity not by itself turn a general Euler method for

$$y' = f(x, y)$$

into a classical Riemann sum for a known function of x alone?

106. In the case

$$y'(x) = g(x),$$

under what conditions on the partition and sample points does

$$\sum_{i=0}^{n-1} g(x_i) \Delta x_i$$

converge to

$$\int_a^b g(x) dx?$$

State the connection with the definition of Riemann integrability.

107. If

$$y'(x) = f(x, y)$$

does depend on y , what breaks in the attempt to interpret Euler's method as an ordinary Riemann sum in the variable x alone? Which additional unknown quantity is being approximated at every step?

108. Compare these two viewpoints:

$$\int_a^b g(x) dx \quad \text{and} \quad y(b) - y(a) \quad \text{for} \quad y' = f(x, y).$$

In what sense are both global accumulations of local data, and in what sense is the second problem structurally more implicit?

109. If two solutions of

$$y' = g(x)$$

have different initial values, why does their difference remain constant in x ? Why is the analogous statement usually false for a general ODE

$$y' = f(x, y)?$$

21 The Connection Between the Net Change Theorem, Differential Equations, and Euler's Method

The connection between the net change theorem, differential equations, and Euler's method is structural. They are different views of the same fundamental operation:

- Differential equations describe *local rates of change*.
- Integration accumulates those local changes into a *global change*.
- Euler's method numerically performs that accumulation step-by-step.

The net change theorem is the bridge connecting these ideas. Suppose that

$$F'(t) = f(t).$$

Then the net change theorem states:

$$F(b) - F(a) = \int_a^b f(t) dt \quad \text{or} \quad F(b) = F(a) + \int_a^b f(t) dt.$$

The interpretation is:

- $f(t)$ is the instantaneous rate of change,
- the integral accumulates infinitesimal changes,
- the result is the total change in F .

Now consider a differential equation:

$$y'(t) = f(t, y(t)).$$

This equation states that the instantaneous change of the unknown function y is determined by the function f .

Applying the net change theorem to the unknown solution y , we integrate both sides from t_0 to t :

$$y(t) - y(t_0) = \int_{t_0}^t f(s, y(s)) ds.$$

If the initial value is

$$y(t_0) = y_0,$$

then:

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds.$$

This equation has a direct interpretation:

- Start with the initial value y_0 ,
- accumulate all infinitesimal changes $f(s, y(s)) ds$,
- obtain the final value $y(t)$.

Thus, a differential equation is fundamentally an accumulation law.

Euler's Method as Numerical Integration

Euler's method arises by approximating the integral with a Riemann sum. Take a small step size h . Over the interval $[t_n, t_n + h]$,

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_n+h} f(s, y(s)) \, ds.$$

Euler's approximation assumes that the slope remains approximately constant over this short interval:

$$f(s, y(s)) \approx f(t_n, y_n).$$

Therefore,

$$\int_{t_n}^{t_n+h} f(s, y(s)) \, ds \approx h f(t_n, y_n).$$

Substituting this approximation gives Euler's update rule:

$$y_{n+1} = y_n + h f(t_n, y_n).$$

This is not merely inspired by integration. It *is* numerical integration. Each Euler step performs the following process:

1. Measure the current rate of change,
2. assume it remains roughly constant over a short interval,
3. accumulate the resulting small change,
4. repeat.

Thus, every Euler step is a discrete version of the net change theorem:

$$\text{new value} = \text{old value} + \text{accumulated change}.$$

This explains why:

- numerical ODE solving is often called *integration*,
- solution curves are called *integral curves*,
- ODE solvers are often called *integrators*.

22 Cauchy Formula for Repeated Integration

22.1 Basic understanding

1. What problem does the Cauchy formula for repeated integration solve?
2. What is meant by the " n -th repeated integral" of a function?
3. Write the definition of the n -fold repeated integral of a function f with the base point a :

$$f^{(-n)}(x) = ?$$

4. Under what conditions on f does the formula hold?
5. What is the main result of the Cauchy formula for repeated integration? State it explicitly.

22.2 Formula and structure

6. Rewrite the repeated integral as a single integral:

$$f^{(-n)}(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt.$$

7. What is the role of the factor $\frac{1}{(n-1)!}$?
8. Why does the kernel $(x-t)^{n-1}$ appear in the formula?
9. What happens in the special case $n = 1$? Verify that the formula reduces correctly.
10. Compute explicitly $f^{(-2)}(x)$ using both:
 - the definition (nested integrals),
 - the Cauchy formula,and verify that they agree.

22.3 Proof understanding

11. What proof technique is used to prove the formula?
12. What is the base case of the induction?
13. Show that for $n = 1$:

$$f^{(-1)}(x) = \int_a^x f(t) dt.$$

14. What assumption is made in the induction step?
15. What must be proven in the induction step?
16. Where is the Leibniz integral rule used in the proof?
17. Compute:

$$\frac{d}{dx} \left[\frac{1}{n!} \int_a^x (x-t)^n f(t) dt \right]$$

and explain why it gives the desired recursive relation.

18. How does the induction step transform an $(n+1)$ -fold integral into the desired single-integral form?
19. Why does the induction argument prove the formula for all $n \in \mathbb{N}$?

22.4 Conceptual understanding

20. Why can repeated integration be "compressed" into a single integral?
21. How is this formula related to convolution?
22. Interpret the kernel $(x - t)^{n-1}$ geometrically or intuitively.
23. How does this formula simplify practical computations?
24. In what situations is this formula especially useful?

22.5 Generalizations

25. How can the formula be extended to non-integer orders of integration?
26. What replaces the factorial $(n - 1)!$ in the generalized version?
27. State the fractional integral version:
$$(J^\alpha f)(x) = ?$$
28. What condition must α satisfy?
29. What is the name of this generalization?
30. How does the formula relate to fractional derivatives (when $\alpha < 0$)?

22.6 Connections and further ideas

31. How is this formula connected to fractional calculus?
32. What is a differintegral?
33. How can repeated integration be used to define differentiation of non-integer order?
34. How is this formula generalized to higher dimensions?
35. What is the Riesz potential?

22.7 Practice problems

36. Compute:

$$\int_0^x \int_0^{t_1} t_2 dt_2 dt_1$$

using the Cauchy formula.

37. Compute the n -fold integral of $f(t) = 1$.
38. Compute the n -fold integral of $f(t) = t^k$.
39. Show that:

$$(J^n f)(x) = (f * g_n)(x)$$

for an appropriate kernel g_n .

40. Verify the formula numerically for a simple function (e.g. $f(t) = e^t$).

23 Differentiation Under the Integral Sign

1. What is the statement of Leibniz's rule for differentiating under the integral sign with constant limits?
2. What is the general form of Leibniz's rule when the limits of integration depend on the parameter?
3. What conditions must be satisfied to justify differentiation under the integral sign (continuity, uniform convergence, dominated convergence, etc.)?
4. What role does continuity of the integrand and its partial derivative play in applying differentiation under the integral sign?
5. How does the Dominated Convergence Theorem justify differentiation under the integral sign?
6. When can uniform convergence be used to justify differentiation under the integral sign?
7. What is the difference between pointwise convergence and uniform convergence in this context?
8. How do you compute $\frac{d}{d\alpha} \int_a^b f(x, \alpha), dx$?
9. How do you compute $\frac{d}{d\alpha} \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha), dx$?
10. What additional terms appear when the limits of integration depend on the parameter?
11. How do you apply differentiation under the integral sign to evaluate difficult integrals?
12. What is the idea behind Feynman's technique for evaluating integrals using parameters?
13. How can introducing a parameter simplify the evaluation of an integral?
14. Give an example where differentiating under the integral sign transforms an integral into a simpler one.
15. How do you recover the original integral after differentiating with respect to a parameter?
16. What initial or boundary conditions are needed after integrating back with respect to the parameter?
17. How do you handle constants of integration when using this method?
18. When is it valid to interchange differentiation and integration?
19. What are common pitfalls when applying differentiation under the integral sign?
20. How does this technique apply to improper integrals?
21. What extra care is needed when the interval of integration is infinite?
22. How do you justify differentiation under the integral sign for $\int_0^\infty f(x, \alpha), dx$?
23. Can differentiation under the integral sign be applied multiple times? Under what conditions?
24. What is the relationship between differentiation under the integral sign and parameter-dependent integrals?
25. How does this method connect to solving differential equations?
26. How can this technique be used in probability theory (e.g., moment generating functions)?
27. How does differentiation under the integral sign relate to Fourier transforms or Laplace transforms?
28. What are some classic integrals that are evaluated using this method?
29. How would you construct your own parameterized integral to apply this technique?
30. How do you verify that your final answer is correct after applying the method?

24 The Gamma Function

1. What is the definition of the Gamma function $\Gamma(x)$ for $x > 0$?
2. For which values of x is the integral definition of $\Gamma(x)$ valid?
3. Show that $\Gamma(1) = 1$.
4. Prove the functional equation $\Gamma(x+1) = x\Gamma(x)$.
5. How does the Gamma function extend the factorial function? State the relationship between $\Gamma(n)$ and $(n-1)!$ for $n \in \mathbb{N}$.
6. Compute $\Gamma(2)$, $\Gamma(3)$, and $\Gamma(4)$ using the functional equation.
7. What is the value of $\Gamma\left(\frac{1}{2}\right)$?
8. How can the integral $\int_0^\infty e^{-t^2} dt$ be used to compute $\Gamma\left(\frac{1}{2}\right)$?
9. What substitution transforms the Gaussian integral into the Gamma function integral?
10. What is the reflection formula for the Gamma function?
11. Use the reflection formula to compute $\Gamma\left(\frac{1}{2}\right)$.
12. What is the duplication formula (Legendre's formula) for the Gamma function?
13. What is the behavior of $\Gamma(x)$ near $x = 0$?
14. Does the Gamma function have any poles? If so, where are they located?
15. Is the Gamma function defined for negative non-integer values? What happens at negative integers?
16. What is the logarithmic derivative of the Gamma function called?
17. Define the digamma function $\psi(x)$.
18. What is the integral representation of $\Gamma(x)$ involving $e^{-t}t^{x-1}$?
19. What is an alternative integral representation of $\Gamma(x)$ using a limit (Euler's limit form)?
20. What is Stirling's approximation for $\Gamma(x)$ for large x ?
21. How can Stirling's formula be used to approximate $n!$?
22. What is the Beta function $B(x, y)$?
23. How is the Beta function related to the Gamma function?
24. Show that $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
25. What is the integral definition of the Beta function?
26. How can you derive the Beta-Gamma relationship using a change of variables?
27. What is the asymptotic growth rate of $\Gamma(x)$ as $x \rightarrow \infty$?
28. Is the Gamma function convex or concave on $(0, \infty)$?
29. What is the logarithmic convexity (log-convexity) property of $\Gamma(x)$?
30. State Bohr-Mollerup theorem and explain its significance.
31. How does the Gamma function behave under scaling transformations?
32. What is the value of $\Gamma\left(\frac{3}{2}\right)$?
33. Compute $\Gamma\left(\frac{5}{2}\right)$ using recursion.

34. Express $\Gamma\left(n + \frac{1}{2}\right)$ in terms of factorials and $\sqrt{\pi}$.
35. What is the relation between $\Gamma(x)$ and improper integrals in probability theory?
36. How does the Gamma function appear in the definition of the Gamma distribution?
37. What is the probability density function of the Gamma distribution?
38. What role does $\Gamma(x)$ play in normalizing probability distributions?
39. How can $\Gamma(x)$ be analytically continued to complex numbers?
40. Is $\Gamma(x)$ an entire function? Why or why not?
41. Where are the zeros of the Gamma function located?
42. What is the Weierstrass product representation of $\Gamma(x)$?
43. How does $\Gamma(x)$ relate to complex analysis and contour integrals?
44. What is Euler's constant γ and how does it relate to $\Gamma(x)$?
45. What is the series expansion of $\log \Gamma(x)$ near $x = 1$?
46. How is the Gamma function used in evaluating definite integrals?
47. Give an example of an integral that evaluates to a Gamma function.
48. How does the substitution $t = -\ln u$ relate the Gamma function to integrals over $(0, 1)$?
49. What is the connection between $\Gamma(x)$ and factorial moments?
50. Why is the Gamma function important in physics and engineering?

25 Derivative of the Inverse Function

1. What is the definition of the inverse of a function f ? Under what condition does an inverse exist?
2. What does it mean for a function to be one-to-one (injective), and why is this property necessary for the existence of an inverse?
3. State the relationship between a function and its inverse using composition.
4. If $y = f(x)$, how do you express x in terms of f^{-1} ?
5. What is the geometric relationship between the graphs of f and f^{-1} ?
6. How can you determine whether a function has an inverse using the horizontal line test?
7. State the formula for the derivative of an inverse function:

$$(f^{-1})'(x) = ?$$

8. Derive the formula for $(f^{-1})'(x)$ starting from the identity $f(f^{-1}(x)) = x$.
9. If $f(a) = b$, express $(f^{-1})'(b)$ in terms of $f'(a)$.
10. Under what conditions on f does the derivative of the inverse function exist?
11. Why must $f'(a) \neq 0$ for the inverse function to be differentiable at $b = f(a)$?
12. Compute $(f^{-1})'(x)$ when $f(x) = x^3 + x$.
13. Let $f(x) = \sin x$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. What is $f^{-1}(x)$ and its derivative?
14. Find the derivative of $\arctan(x)$ using the inverse function derivative formula.
15. Show that the derivative of $\ln x$ can be derived from the inverse of e^x .
16. If $f'(x)$ is known, describe the steps to compute $(f^{-1})'(x)$ at a specific point.
17. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very large? What is the geometric interpretation?
18. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very small (but nonzero)?
19. Explain why the slopes of inverse functions at corresponding points are reciprocals.
20. If f is decreasing and invertible, what can you say about the sign of $(f^{-1})'(x)$?
21. Let $f(x) = x^5$. Compute $(f^{-1})'(x)$ explicitly.
22. If $f(x) = \tan x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, compute the derivative of its inverse.
23. Verify that $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$ using the inverse function rule.
24. How does implicit differentiation relate to finding the derivative of an inverse function?
25. If f is differentiable and strictly monotonic on an interval, what can be said about the differentiability of f^{-1} ?
26. Give an example of a function that is invertible but whose inverse is not differentiable at some point.
27. What is the domain of f^{-1} in terms of the range of f ?
28. What is the range of f^{-1} in terms of the domain of f ?
29. How can you use the inverse function derivative formula to avoid explicitly finding the inverse?
30. Let $f(2) = 5$ and $f'(2) = 3$. Find $(f^{-1})'(5)$.

26 Newton's (Newton–Raphson) Method

1. What is the goal of a root-finding algorithm?
2. State the Newton method iteration formula for finding a root of a function $f(x)$.
3. How do you obtain the next iterate x_{n+1} from the tangent line?
4. Derive the Newton iteration by applying a first-order Taylor expansion of $f(x)$ around x_n .
5. What assumptions about $f(x)$ are required for Newton's method to work?
6. What is meant by an initial guess x_0 , and why is it important?
7. Define what it means for Newton's method to converge.
8. What is quadratic convergence?
9. Why does Newton's method converge faster near the root than far from it?
10. What happens if $f'(x_n) = 0$ during the iteration?
11. Why is Newton's method considered a local method?
12. Why can Newton's method fail to converge for some initial guesses?
13. Give an example of a function and initial guess where Newton's method diverges.
14. What is meant by the basin of attraction of a root?
15. How does the choice of initial guess affect which root Newton's method converges to?
16. What is a multiple root, and how does it affect convergence of Newton's method?
17. How can Newton's method be modified to handle multiple roots?
18. What stopping criteria can be used to terminate Newton's method?
19. Compare stopping criteria based on $|x_{n+1} - x_n|$ and $|f(x_n)|$.
20. How is Newton method formulated in terms of a vector-valued function $F(x)$?
21. How can Newton's method be extended to systems of nonlinear equations?
22. How do we define the Jacobian matrix $J_F(x)$ for a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$?
23. Write the Newton iteration formula for solving $F(x) = 0$ in $\mathbb{R}^n$.
24. What role does the Jacobian matrix play in multidimensional Newton's method?
25. What issues arise when the Jacobian is singular or nearly singular?
26. What is a damped (or modified) Newton method?
27. Why might damping improve convergence?
28. How can Newton's method be used to find extrema of a function?
29. How is Newton's method related to solving $f'(x) = 0$?
30. What is the difference between Newton's method for optimization and for root finding?
31. What is the order of convergence of Newton's method compared to secant and bisection methods?
32. What is the secant method, and how does it approximate Newton's method?
33. When might the secant method be preferred over Newton's method?
34. How can automatic differentiation help in implementing Newton's method?

35. What is line search in the context of Newton's method?
36. How can Newton's method behave near inflection points?
37. What happens if the function is not differentiable at the root?
38. Why can Newton's method oscillate between points instead of converging?
39. How can you detect divergence in practice?
40. What are hybrid methods that combine Newton's method with bracketing methods?
41. Why are hybrid methods often preferred in practice?
42. What is the role of scaling in Newton's method for systems?
43. What is meant by the error $e_n = x_n - r$, and how does it evolve in Newton's method?
44. Derive the error recurrence relation for Newton's method near a simple root.
45. What is the goal of Newton's method in the context of optimization?
46. How do we derive Newton's method for finding a local minimum of a scalar function $f(x)$?
47. What role does the gradient $\nabla f(x)$ play in Newton's method?
48. What is the Hessian matrix $\nabla^2 f(x)$ and how is it defined?
49. Write the Newton iteration formula for optimization in $\mathbb{R}^n$.
50. How does Newton's method extend to complex-valued functions?
51. What are basins of attraction in the complex plane?
52. How can we visualize the convergence behavior of Newton's method in the complex plane?
53. How do Newton fractals arise from iterating Newton's method in $\mathbb{C}$?
54. Why do the boundaries between basins of attraction exhibit fractal structure?
55. What are fixed points of the Newton iteration map?
56. Are all fixed points of the Newton map roots of $f(z)$?
57. What are attracting, repelling, and neutral fixed points in this context?

27 Roots of Unity

1. What is meant by an n th root of unity?
2. Write the equation whose solutions are the n th roots of unity.
3. How many distinct n th roots of unity are there? Why?
4. Express the n th roots of unity in exponential form using Euler's formula.
5. Write the general formula for the k th n th root of unity.
6. What is the principal (or primitive) n th root of unity?
7. How can all n th roots of unity be generated from a single primitive root?
8. What is the difference between a root of unity and a primitive root of unity?
9. State the condition under which a root of unity is primitive.
10. How many primitive n th roots of unity are there? Which function counts them?
11. What is Euler's totient function $\varphi(n)$ and how does it relate to roots of unity?
12. Represent the n th roots of unity geometrically in the complex plane.
13. What is the magnitude and argument of each n th root of unity?
14. How are the n th roots of unity distributed on the complex unit circle?
15. What is the angle between consecutive n th roots of unity?
16. Compute the sum of all n th roots of unity. What is the result?
17. Why does the sum of all n th roots of unity equal zero?
18. Compute the product of all n th roots of unity.
19. What symmetry properties do the n th roots of unity exhibit?
20. Show that the n th roots of unity form a group under multiplication.
21. What type of group is formed by the n th roots of unity?
22. What is the identity element in this group?
23. What is the inverse of a given n th root of unity?
24. Show that the set of n th roots of unity is cyclic.
25. What is a generator of the group of n th roots of unity?
26. For which values of k is ζ^k a primitive root, where ζ is a primitive n th root?
27. Solve the equation $z^n = 1$ explicitly for small values such as $n = 2, 3, 4$.
28. What are the cube roots of unity? Write them explicitly.
29. What are the fourth roots of unity? Write them explicitly.
30. How can roots of unity be used to factor the polynomial $x^n - 1$?
31. Write the factorization of $x^n - 1$ in terms of its complex roots.
32. What are cyclotomic polynomials and how are they related to roots of unity?
33. Define the n th cyclotomic polynomial $\Phi_n(x)$.
34. How does $x^n - 1$ factor into cyclotomic polynomials?
35. What is the relationship between primitive roots of unity and cyclotomic polynomials?

36. How can roots of unity be used to evaluate sums such as $\sum_{k=0}^{n-1} \zeta^k$?
37. Evaluate $\sum_{k=0}^{n-1} \zeta^{mk}$ for an integer m .
38. Under what condition does $\sum_{k=0}^{n-1} \zeta^{mk} = 0$?
39. What happens when m is a multiple of n in the above sum?
40. How are roots of unity used in discrete Fourier transforms (DFT)?
41. What is the connection between roots of unity and periodicity?
42. How can roots of unity simplify computations involving periodic sums?
43. Show that if ζ is an n th root of unity, then $\bar{\zeta} = \zeta^{-1}$.
44. What is the complex conjugate of a root of unity geometrically?
45. How do roots of unity relate to rotations in the complex plane?
46. What is the minimal polynomial of a primitive n th root of unity?
47. How does the degree of $\Phi_n(x)$ relate to $\varphi(n)$?
48. In what fields do primitive n th roots of unity lie?
49. What is the significance of roots of unity in number theory?
50. How are roots of unity used in solving polynomial equations?

28 Complex Numbers as Matrices / Representation Theory

<https://www.youtube.com/watch?v=hsveVFoIJPM>

1. What does it mean to represent an abstract algebraic object (like a group element or number) as a matrix?
2. What is the general idea of a **representation** in the context of linear algebra and group theory?
3. How can a complex number $a + bi$ be represented as a 2×2 real matrix?
4. Write explicitly the matrix corresponding to the complex number $a + bi$.
5. Show how matrix addition corresponds to addition of complex numbers under this representation.
6. Show how matrix multiplication corresponds to multiplication of complex numbers under this representation.
7. Why is this matrix representation of complex numbers structure-preserving?
8. What algebraic structure is preserved when mapping complex numbers to matrices?
9. What property must a mapping satisfy to be considered a homomorphism?
10. Verify that the mapping from complex numbers to matrices is a homomorphism.
11. What is the identity element in the complex numbers, and what matrix represents it?
12. What matrix corresponds to the imaginary unit i ?
13. Compute the square of the matrix representing i and interpret the result.
14. How does this matrix representation encode the fact that $i^2 = -1$?
15. What geometric transformation in $\mathbb{R}^2$ corresponds to multiplication by i ?
16. How does multiplication by a general complex number $a + bi$ act geometrically on the plane?
17. How is rotation represented in terms of matrices?
18. How does scaling appear in the matrix representation of a complex number?
19. Explain how complex multiplication combines rotation and scaling.
20. What is the determinant of the matrix corresponding to $a + bi$?
21. What is the geometric meaning of this determinant?
22. What is the trace of the matrix representation of $a + bi$?
23. How does the modulus $|a + bi|$ relate to the determinant of the matrix?
24. What condition on a and b makes the matrix invertible?
25. What is the matrix corresponding to the inverse of a complex number?
26. Show that the inverse matrix corresponds to the multiplicative inverse of the complex number.
27. How does this representation help visualize complex numbers as linear transformations?
28. What is the connection between this representation and rotations in the plane?
29. How can this idea be generalized to represent other algebraic structures using matrices?
30. What is a group representation in general?
31. Why are matrices particularly useful for representing abstract algebraic objects?
32. How does this example motivate the study of representation theory?

33. What advantages do matrix representations provide for computation?
34. How can eigenvalues of these matrices be interpreted in terms of complex numbers?
35. What happens when you diagonalize the matrix corresponding to a complex number?
36. How does this representation connect linear algebra with complex analysis?
37. In what sense are complex numbers “the same” as a subset of 2×2 real matrices?
38. What is the dimension of the vector space in which these matrices act?
39. How does this perspective change your understanding of multiplication by complex numbers?
40. Can every linear transformation of $\mathbb{R}^2$ be represented by a complex number? Why or why not?
41. What distinguishes matrices that correspond to complex numbers from arbitrary 2×2 matrices?
42. How would you test whether a given 2×2 matrix corresponds to a complex number?
43. What deeper insight does this example give about the relationship between algebra and geometry?

29 Hidden Constants in Integration by Parts

<https://www.youtube.com/watch?v=N5xtRr0jfWI>

<https://www.youtube.com/watch?v=V8uZYdd4Jnk>

1. Why does every indefinite integral include an arbitrary constant C ?

2. Derive the integration by parts formula from the product rule.

3. Why is the equation

$$\int u \, dv = uv - \int v \, du$$

technically incomplete if we ignore constants of integration? Where are the hidden constants in this equation?

4. Why can ignoring arbitrary constants sometimes lead to paradoxes or false identities when manipulating indefinite integrals?

5. Consider this example

$$I = \int \underbrace{\frac{f'(x)}{f(x)}}_{uv'} dx = \underbrace{\frac{f(x)}{f(x)}}_{uv} - \int \underbrace{-\frac{f'(x)}{f^2(x)}}_{u'} \underbrace{f(x)}_v dx = 1 + \int \frac{f'(x)}{f(x)} dx = 1 + I$$

$$\begin{aligned} u &= \frac{1}{f(x)} & v' &= f'(x) \\ u' &= -\frac{f'(x)}{f(x)^2} & v &= f(x) \end{aligned}$$

Where is the error in this paradoxical derivation?

6. Consider this example

$$I = \int f(x)g'(x) \, dx$$

$$\begin{aligned} u &= f(x) & v' &= g'(x) \\ u' &= f'(x) & v &= g(x) + A \end{aligned}$$

Show how all the constants of integration are absorbed into one when applying integration by parts.

7. Why can repeated applications of integration by parts accumulate multiple constants?

8. How can these multiple constants always be absorbed into a single constant in the final answer?

9. Why is it sometimes useful to deliberately choose a non-zero constant when computing an antiderivative inside integration by parts?

10. If the integration by part is applied naively to integral $\int x \arctan(x) \, dx$ it leads to an incomplete solution

$$\int \underbrace{x}_{v'} \underbrace{\arctan(x)}_u dx = \underbrace{\frac{x^2}{2}}_v \underbrace{\arctan(x)}_u - \int \underbrace{\frac{x^2}{2}}_v \underbrace{\frac{1}{x^2+1}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

11. If the integration by part is applied naively to integral $\int \ln(x+2) \, dx$ it leads to an incomplete solution

$$\int \underbrace{1}_{v'} \underbrace{\ln(x+2)}_u dx = \underbrace{x}_v \underbrace{\ln(x+2)}_u - \int \underbrace{x}_v \underbrace{\frac{1}{x+2}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

12. Why are definite integrals immune to the constant-of-integration ambiguity?

30 Geometric Meaning of the Third Derivative

<https://www.youtube.com/watch?v=SovllrJUQ64>

1. What is the geometric meaning of the first derivative $f'(x)$ in terms of the graph of $f(x)$?
2. How can you interpret the second derivative $f''(x)$ geometrically?
3. What does it mean for $f''(x) > 0$ in terms of the shape of the graph?
4. What does it mean for $f''(x) < 0$?
5. Define an inflection point in terms of concavity.
6. What condition on $f''(x)$ is necessary for an inflection point?
7. Why is $f''(x) = 0$ not sufficient to guarantee an inflection point?
8. What additional condition ensures that a point where $f''(x) = 0$ is actually an inflection point?
9. How can sign changes of $f''(x)$ be used to detect inflection points?
10. What is the geometric meaning of the third derivative $f'''(x)$?
11. How can $f'''(x)$ be interpreted as a rate of change of another geometric quantity?
12. If $f'''(x) > 0$, what does this say about how concavity is changing?
13. If $f'''(x) < 0$, what does this imply about the change in concavity?
14. How does $f'''(x)$ describe the “bending behavior” of a curve beyond concavity?
15. Explain how the third derivative relates to the steepness of the tangent slope.
16. How does the graph of $f'(x)$ help visualize $f''(x)$?
17. How does the graph of $f''(x)$ help visualize $f'''(x)$?
18. If $f''(x)$ has a local maximum or minimum, what can you say about $f'''(x)$ at that point?
19. What is the relationship between $f'''(x) = 0$ and extrema of $f''(x)$?
20. Can $f'''(x) = 0$ correspond to an inflection point of $f(x)$? Explain.
21. How would you distinguish between:
 - an inflection point of $f(x)$
 - an extremum of $f''(x)$
22. Describe how successive derivatives correspond to successive “rates of change” geometrically.
23. Give a physical interpretation of $f'(x)$, $f''(x)$, and $f'''(x)$ in motion (position, velocity, acceleration, ...).
24. What is the physical meaning of the third derivative in kinematics (often called jerk)?
25. How does the sign of the third derivative affect motion in a physical system?
26. Construct an example function where $f''(x) = 0$ but there is no inflection point.
27. Construct an example where $f'''(x) = 0$ but $f''(x)$ does not change sign.
28. Given a graph, how would you visually estimate where $f'''(x)$ is positive or negative?
29. How can you use higher derivatives to understand increasingly subtle features of a curve?
30. Why is the third derivative rarely discussed compared to the first and second in basic calculus?
31. In what types of problems does understanding the third derivative become important?

31 Second Derivative Test and Hessian

<https://www.youtube.com/watch?v=z0bSdV8DA4M>

31.1 Core Concept: What is the second derivative test really doing?

- What does the second derivative test attempt to determine about a function at a critical point?
- Why is checking only the first derivative insufficient for classifying critical points?
- In one variable, what does the sign of $f''(x)$ tell us geometrically?
- How can you interpret $f''(x)$ in terms of curvature?

31.2 From 1D to Multivariable Functions

- What is the analogue of the second derivative for multivariable functions?
- What is the Hessian matrix, and how is it constructed?
- Why can't we just look at "the second derivative" in multiple dimensions?
- What does the Hessian represent geometrically?

31.3 Quadratic Approximation

- What is the second-order Taylor expansion of a function near a critical point?
- Why does the linear term vanish at a critical point?
- What role does the quadratic term play in determining local behavior?
- How does the function locally resemble a quadratic form?

31.4 Quadratic Forms and Geometry

- What is a quadratic form?
- How can the expression $\mathbf{x}^T H \mathbf{x}$ describe local curvature?
- What does it mean for a quadratic form to be:
 - Positive definite?
 - Negative definite?
 - Indefinite?

31.5 Classification of Critical Points

- How does the definiteness of the Hessian classify a critical point?
- What condition corresponds to a local minimum?
- What condition corresponds to a local maximum?
- What condition corresponds to a saddle point?
- What happens if the Hessian is degenerate (determinant zero)?

31.6 Eigenvalues Interpretation

- Why are eigenvalues the key to understanding the Hessian?
- How do eigenvalues relate to curvature along different directions?
- What does it mean if all eigenvalues are positive?
- What does it mean if eigenvalues have mixed signs?
- How do eigenvectors relate to principal directions of curvature?

31.7 Geometric Intuition

- Why is a saddle point associated with “curving up in some directions and down in others”?
- How can you visualize curvature along different directions through a point?
- Why is the second derivative test really a statement about curvature in all directions?

31.8 Connection to Linear Algebra

- Why can any symmetric matrix be diagonalized?
- How does diagonalizing the Hessian simplify the quadratic form?
- What does the diagonal form reveal about the function locally?

31.9 Deeper Understanding

- Why is the second derivative test fundamentally about approximating the function by a quadratic surface?
- In what sense is the test coordinate-independent?
- How does this perspective generalize beyond $\mathbb{R}^2$ or $\mathbb{R}^3$?

31.10 Edge Cases and Limitations

- Why does the test fail when eigenvalues are zero?
- What additional analysis is needed in degenerate cases?
- Can higher-order terms change the classification?

32 Frullani's Integral

1. State Frullani's integral theorem.
2. What conditions on $f(x)$ are typically required for Frullani's theorem to hold?
3. Why must the limits $f(0)$ and $f(\infty)$ exist?
4. What is the role of the condition $a, b > 0$?
5. Why does the logarithm naturally appear in Frullani-type integrals?
6. Derive

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \ln \frac{b}{a}.$$

7. Explain why

$$\int_0^\infty \frac{e^{-ax}}{x} dx$$

diverges.

8. Why does subtracting two individually divergent integrals sometimes produce a convergent result?
9. What happens if $a = b$?
10. What happens if one of the parameters tends to zero?
11. What happens if one of the parameters tends to infinity?
12. Show how Frullani's theorem can be derived using differentiation under the integral sign.
13. Define

$$F(a) = \int_0^\infty \frac{e^{-x} - e^{-ax}}{x} dx.$$

Compute $F'(a)$, use the initial condition $F(1) = 0$ to determine the constant of integration, and write down the closed form for $F(a)$.

14. When applying the Feynman trick to

$$I(a) = \int_0^\infty \frac{f(x) - f(ax)}{x} dx,$$

what initial condition pins down the constant of integration? Why does this condition hold?

15. Differentiate

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$$

with respect to a , solve the resulting equation, and verify that the answer matches Frullani's theorem.

16. In the Feynman approach, differentiating with respect to a removes the $\frac{1}{x}$ factor from the integrand. Why does this make the integral tractable, and what condition on f justifies differentiating under the integral sign?
17. Why does the final answer depend only on $f(0)$ and $f(\infty)$ rather than the full shape of f ?
18. Give an example of a function for which Frullani's theorem fails, and identify which hypothesis is violated.
19. Derive Frullani's theorem from the identity

$$f(ax) - f(bx) = \int_b^a x f'(tx) dt.$$

20. Compute

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx.$$

21. Compute

$$\int_0^\infty \frac{\ln(1+ax) - \ln(1+bx)}{x} dx.$$

Verify the answer by checking the $a = b$ limit.

22. Does Frullani's theorem apply to $\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx$? Identify which hypothesis is (or is not) satisfied, and compute the integral by another method.

23. What changes if the integration interval is $(0, T)$ rather than $(0, \infty)$?

24. Write $\frac{1}{x} = \int_0^\infty e^{-xs} ds$ and substitute into $\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$. After a formal interchange of the order of integration, express the result in terms of the Laplace transform $\mathcal{L}\{f\}(s) = \int_0^\infty f(x)e^{-sx} dx$.

25. Show that

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \int_0^\infty \left[\frac{1}{a} \mathcal{L}\{f\}\left(\frac{s}{a}\right) - \frac{1}{b} \mathcal{L}\{f\}\left(\frac{s}{b}\right) \right] ds.$$

Verify this for $f(x) = e^{-x}$ by computing the right-hand side directly and confirming the answer equals $\ln(b/a)$.

26. State the identity

$$\int_0^\infty \frac{g(x)}{x} dx = \int_0^\infty \mathcal{L}\{g\}(s) ds$$

and give sufficient conditions for its validity. Use it to rederive Frullani's theorem in one step by setting $g(x) = f(ax) - f(bx)$ and computing $\mathcal{L}\{g\}(s)$ explicitly.

27. In the Laplace-transform approach, why is the condition $g(0) = 0$ necessary for the identity in the previous question? Show that $g(0) = 0$ follows automatically when $g(x) = f(ax) - f(bx)$ and the hypotheses of Frullani's theorem are satisfied.

28. What justifies the interchange of the order of integration in the Laplace-transform proof? State the theorem you invoke (e.g. Tonelli's theorem) and verify its hypotheses for $f(x) = e^{-x}$.

29. How is Frullani's theorem connected to Laplace transforms in a broader sense? (*Hint*: consider the formula $\int_0^\infty f(x)/x dx = \int_0^\infty F(s) ds$, where $F = \mathcal{L}\{f\}$, and how it applies once the condition $g(0) = 0$ is ensured.)

30. Both the Feynman trick and the Laplace-kernel method deal with the $\frac{1}{x}$ factor. In the Feynman approach, $\frac{1}{x}$ is absorbed by differentiating with respect to a parameter; in the Laplace approach, $\frac{1}{x}$ is represented as $\int_0^\infty e^{-xs} ds$. Explain the structural analogy between these two approaches, and describe a setting in which one is preferable to the other.

31. What assumptions are required to differentiate under the integral sign? State a sufficient condition (e.g. dominated convergence) and verify it for $f(x) = e^{-x}$ in the Feynman proof of Frullani's theorem.

33 Dirichlet's Integral

1. State Dirichlet's integral:

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What is its value?

2. Why is the integrand not actually singular at $x = 0$? Compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

3. Does

$$\int_0^\infty \left| \frac{\sin x}{x} \right| dx$$

converge? What does this tell you about the convergence of Dirichlet's integral?

4. Use integration by parts on

$$\int_1^R \frac{\sin x}{x} dx$$

to show that the limit as $R \rightarrow \infty$ exists.

5. For $a > 0$, compute

$$\int_0^\infty \frac{\sin(ax)}{x} dx$$

by the substitution $u = ax$.

6. What happens to the previous integral when $a = 0$?

7. What happens when $a < 0$? Use the oddness of $\sin$ to determine the sign.

8. Compare

$$\int_0^\infty \frac{\sin(ax)}{x} dx \quad \text{and} \quad \int_0^\infty \sin(ax) dx.$$

Why does the factor $\frac{1}{x}$ matter so much?

9. Define

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx, \quad a > 0.$$

Why is this integral easier to handle than

$$\int_0^\infty \frac{\sin x}{x} dx$$

directly?

10. Differentiate $F(a)$ with respect to a and show that

$$F'(a) = - \int_0^\infty e^{-ax} \sin x dx.$$

11. Compute

$$\int_0^\infty e^{-ax} \sin x dx$$

for $a > 0$.

12. Deduce that

$$F'(a) = -\frac{1}{1+a^2}.$$

Integrate this with respect to a .

13. Use the condition

$$\lim_{a \rightarrow \infty} F(a) = 0$$

to determine the constant of integration.

14. Show that

$$F(a) = \frac{\pi}{2} - \arctan a = \arctan \frac{1}{a}.$$

15. Let $a \rightarrow 0^+$ and recover Dirichlet's integral.

16. Why is introducing the factor e^{-ax} a useful trick before letting $a \rightarrow 0^+$?

17. Verify the identity

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt.$$

18. Substitute this identity into

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What formal double integral do you obtain?

19. Why is it dangerous to interchange the order of integration in that double integral without extra justification?

20. Explain why

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

can be viewed as the Laplace transform of $\frac{\sin x}{x}$ evaluated at a .

21. If

$$f(x) = \frac{\sin x}{x},$$

write $F(a) = \mathcal{L}\{f\}(a)$. What does the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

tell you about $\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$?

22. Starting from

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt,$$

use linearity of the Laplace transform to derive

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \mathcal{L}\{\cos(tx)\}(a) dt.$$

Why is this a safer way to use the identity than working directly with

$$\int_0^\infty \frac{\sin x}{x} dx?$$

23. Compute

$$\mathcal{L}\{\cos(tx)\}(a)$$

for $a > 0$, and use it to show that

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \frac{a}{a^2 + t^2} dt.$$

In other words, show that

$$F(a) = \int_0^1 \int_0^\infty e^{-ax} \cos(tx) dx dt.$$

24. Evaluate

$$\int_0^1 \frac{a}{a^2 + t^2} dt$$

and recover the formula

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \frac{\pi}{2} - \arctan a.$$

25. Use the Laplace-transform identity

$$\mathcal{L}\{f'\}(a) = a\mathcal{L}\{f\}(a) - f(0)$$

with

$$f(x) = \frac{\sin x}{x}$$

to explain informally why the transform of $\frac{\sin x}{x}$ should be related to an arctangent.

26. Why does the limit

$$\lim_{a \rightarrow 0^+} \mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$$

recover Dirichlet's integral?

27. Compare these two viewpoints: differentiating

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

with respect to a , and recognizing $F(a)$ as a Laplace transform from the start. Why are these really two versions of the same idea?

28. Which route feels more natural if you already know Laplace transforms well: differentiating $F(a)$ first, or rewriting $F(a)$ as a Laplace transform of $\frac{\sin x}{x}$? Explain your answer.

29. Why is the value $\pi/2$ plausible from the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

when $a \rightarrow 0^+$?

30. Explain in words why oscillation of $\sin x$ is essential for convergence on $(1, \infty)$.

31. Sketch the graph of $\frac{\sin x}{x}$ and explain how the alternating positive and negative lobes fit the value of the integral.

32. Compute

$$\int_0^\infty \frac{\sin(bx)}{x} dx$$

for $b > 0$, and verify your formula by checking the special case $b = 1$.

34 Cauchy Principal Value

1. What problem is the Cauchy principal value trying to solve?

2. Why does

$$\int_{-1}^1 \frac{1}{x}, dx$$

fail to exist as an ordinary improper integral?

3. What is the difference between an improper integral and a principal value?

4. Why is a principal value not simply another way to evaluate an improper integral?

5. What does the word “symmetric” mean in the definition of principal value?

6. Why is symmetry essential in the definition?

7. Can a principal value exist when the corresponding improper integral does not?

8. Can an improper integral exist when the principal value also exists?

9. What information is lost when we replace an improper integral by a principal value?

10. What geometric intuition can be attached to principal values?

34.1 Computational Questions

11. Compute

$$\text{PV} \int_{-1}^1 \frac{1}{x}, dx.$$

12. Why does the answer not contradict the fact that the ordinary improper integral diverges?

13. Compute

$$\text{PV} \int_{-2}^3 \frac{1}{x}, dx.$$

14. Why is the answer no longer zero?

15. Derive a general formula for

$$\text{PV} \int_a^b \frac{1}{x}, dx,$$

assuming $a < 0 < b$.

16. Compute

$$\text{PV} \int_{-1}^1 \frac{x}{x^2 + 1}, dx.$$

17. Compute

$$\text{PV} \int_{-1}^1 \frac{1+x}{x}, dx.$$

18. Why is it useful to split the integrand into a singular part and a regular part?

19. Compute

$$\text{PV} \int_{-1}^1 \frac{\sin x}{x}, dx.$$

20. Is a principal value always finite when the singularity is of the form $1/x$?

34.2 Questions About Singularities

21. Why does $1/x$ behave differently from $1/x^2$?

22. Does

$$\text{PV} \int_{-1}^1 \frac{1}{x^2}, dx$$

exist?

23. If not, why not?

24. What property of $1/x$ allows cancellation?

25. What property of $1/x^2$ prevents cancellation?

26. For which powers p does

$$\text{PV} \int_{-1}^1 \frac{1}{x^p}, dx$$

exist?

27. What role does oddness or evenness of a function play?

28. When does symmetry force a principal value to be zero?

29. Can an odd function have a divergent principal value?

30. Can an even function have a finite principal value?

34.3 Questions About Infinity

31. What is

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x}, dx?$$

32. How does this differ from

$$\int_{-\infty}^{\infty} \frac{1}{x}, dx?$$

33. Why must limits at $+\infty$ and $-\infty$ be taken symmetrically?

34. What happens if the two limits are taken independently?

35. Give an example where symmetric truncation gives a finite value but ordinary convergence fails.

36. Why is

$$\text{PV} \int_{-\infty}^{\infty} \sin x, dx$$

an interesting example?

37. How does principal value at infinity differ from principal value around an interior singularity?

34.4 Theory Questions

38. State the formal definition of the Cauchy principal value for an interior singularity.

39. State the formal definition for an integral over the whole real line.

40. Why is the principal value defined as a limit?

41. Under what conditions does the principal value agree with the ordinary improper integral?

42. Is the principal value linear?

43. When is it legitimate to split a principal value integral into two pieces?

44. Can principal values always be manipulated algebraically as ordinary integrals?
45. What pitfalls arise when changing variables in principal value integrals?
46. Why must one be careful about symmetry when performing substitutions?
47. Is principal value integration compatible with all standard integration theorems?
48. Which familiar theorems from elementary calculus continue to work unchanged?
49. Which require extra hypotheses?
50. Why do analysts treat principal values as generalized notions of integration?